

Defocus Rephrasing: Right dislocation and syntax-prosody mismatches^{*}

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Abstract

This study adopts a rigorous interface approach to investigate the role of defocus, the systematic lack of focus, in syntax-prosody mapping. Despite the abundance of work on focus prosody, how defocus may affect the mapping and whether its effects should be distinguished from those of focus remains relatively unknown. We argue that defocus can, and should in fact, be separated from focus for prosodic phrasing, drawing crucial support from right dislocation (RD) in Cantonese and Mandarin. In RD, elements are displaced or copied to the right of a sentence, a position that resists semantic, alternative-based focus. We show that although RD consists of two syntactic clauses, it only has one intonational phrase, constituting a syntax-prosody mismatch. We provide evidence from both phonological processes and phonetic cues with acoustic experiments. We propose that defocus cannot receive head prominence and triggers rephrasing to avoid illegitimate headless intonational phrases, leading to the mismatch in RD. Importantly, such a mismatch is not due to any preceding focus, which independently does not trigger rephrasing in both languages. We offer an Optimality-Theoretic account with constraint ranking to capture the asymmetry between focus and defocus. The findings illustrate the intricate interplay of semantic notions with syntax and prosody, and have implications for the modeling of prosodic (non-)prominence.

Keywords: defocus, syntax-prosody mapping, prosodic rephrasing, right dislocation, intonational phrase, prominence

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1 Introduction

1.1 Background

Two central questions in the study of the syntax-phonology interface are how syntactic structures are mapped onto prosodic domains, namely, the constituents in The Prosodic Hierarchy given in (1), and what factors may lead to syntax-prosody mismatches (Selkirk 1978, 1984; Nespor and Vogel 1986; Pierrehumbert and Beckman 1988; L. J. Downing 1998; Ito and Mester 2012, *i.a.*). This study aims at investigating how focus, particularly the *lack* of focus (**defocus**), interacts with syntax-prosody mapping.

(1) The Prosodic Hierarchy (Selkirk 1978)

Utt	Utterance
IP/ ι	intonational phrase
PhP/ ϕ	phonological phrase
PWd/ ω	prosodic word
Ft/ Σ	foot
σ	syllable

The effects of focus on prosody have long been recognized (Jackendoff 1972; Pierrehumbert and Beckman 1988; Truckenbrodt 1995; Büring 2001). Focus often has phonetic effects on the focal field (*focal prominence*), manifested as increased pitch peak and intensity, expanded pitch range, lengthened duration, etc., and on the post-focal field as well (*post-focal reduction*, also called post-focus compression, PFC), manifested as reduced pitch range and/or decreased pitch peak and intensity (Cooper, Eady, and Mueller 1985; Xu 1999; Xu, Chen, and Wang 2012; Liu and Xu 2005). One example is stress shift in English. In (2a), the nuclear stress is by default on the rightmost phrase *Tuesday*. In (2b), focusing the subject *Bill* (marked by the subscripted F) shifts the stress to *Bill*. Moreover, the post-focal phrases in the domain of focus (marked by the subscripted DoF) are deaccented.

(2) Stress shift in English (Truckenbrodt 1995:156)

- a. John met Mary on TUESDAY. (nuclear stress)
- b. (No,) [BILL_F met Mary on Tuesday]_{DoF}. (focal stress)

There are two approaches to interpret these phonetic effects, based on two distinct hypotheses of the role of focus in syntax-prosody mapping, dubbed as the REPHRASING view and the NO-REPHRASING view in this paper. The predominant REPHRASING approach treats the phonetic effects as results from prosodic phrasing (Pierrehumbert and Beckman 1988; Truckenbrodt 1995; Selkirk 2008; Kratzer and Selkirk 2020, *i.a.*). Focus is assumed to be mapped onto prosodic heads, which are in turn aligned with edges. Prosodic rephrasing is triggered when the focus is not properly aligned with edges of prosodic domains, leading to syntax-prosody mismatches. The rephrasing is manifested through boundary insertion and deletion that lead to metrical prominence on the focus (e.g., boosted f0 peak) and post-focal deaccenting.

(3) The REPHRASING view: focus $\rightarrow$ prosodic head

Focus, as least in languages that mark focus by prominence, triggers rephrasing when focus is misaligned with prosodic edges.

The alternative NO-REPHRASING approach, as motivated by recent empirical advances (see, e.g.,

Ishihara 2016 for the lack of boundary insertion in Japanese, also Shinya 1999; Kubozono 2007; and Wagner and McAuliffe 2019; D. Wu 2021; Zhang, Clayards, and Wagner 2024 for the lack of boundary deletion in English and Mandarin Chinese; see also §7), proposes that misaligned focus does not necessarily trigger rephrasing (Féry and Ishihara 2010; Féry 2013; Ishihara 2011, 2016). The phonetic effects of focus are treated as direct manipulation of the f0 register instead of the (indirect) results of prosodic phrasing, unleashing the absolute correspondence between prominence and prosodic heads.

(4) The NO-REPHRASING view: focus $\neq$ prosodic head

Focus, even in languages that mark focus by prominence, does not necessarily trigger rephrasing. Prominence can a result of direct manipulation of pitch register.

These two views, however, only concern whether the *presence* of focus triggers rephrasing and mismatches. Little attention has yet been paid on the *absence* of focus, namely, **defocus** (also called anti-focus, e.g., Molnárfi 2002; T. T.-M. Lee 2020). Different from post-focal elements (which are defined by following a focus), defocus refers to elements at a structural position that systematically resists focus interpretation. To the best of our knowledge, there is no discussion in the literature that addresses whether defocus triggers rephrasing and syntax-prosody mismatches.¹ The major goal of this study is hence to explore the less-understood role of defocus in the syntax-prosody interface. We offer crucial evidence from right dislocation (henceforth RD) in Cantonese and Mandarin, where the dislocated position resists focus interpretation and is always defocused. We argue that defocus can lead to rephrasing, and its role must be distinguished from the role of focus. This view is referred to here as the **DEFOCUS REPHRASING** view. Specifically, we demonstrate that there is a syntax-prosody mismatch that results from the presence of defocus in RD. In the following, we briefly introduce RD in both languages, and then preview our proposal.

1.2 Right dislocation in Cantonese and Mandarin

RD in Cantonese and Mandarin is characterized by displacement or doubling of elements at the end of a sentence, after sentence-final particles (SFPs) when they are present (Chao 1968; Lu 1980; Cheung 1997, 2009; T. T.-M. Lee 2017, *i.a.*). We refer to the elements following SFPs as *right-dislocated (RD) chunks*, and the rest (including the SFPs) as *main chunks*:²

$$(5) \quad \overbrace{[\dots (ZP_i/XP_i) \text{ YP (SFP)}]}^{\text{main chunk}} \quad \overbrace{ZP_i}^{\text{RD chunk}}$$

RD comes in two major types (Cheung 2015; T. T.-M. Lee 2021; Yip 2025a): (i) gapped RD, where the displaced RD chunk corresponds to a gap in the main chunk; (ii) gapless RD, where the RD chunk corresponds to an identical element or a non-identical element (e.g., a resumptive pronoun) in the main chunk. The two types are exemplified by (7) and (8) respectively, with (6) serving as the baseline.³

1. There are a lot of discussions on the prosodic phrasing of given (old information) elements (e.g., Kratzer and Selkirk 2020). It is, however, important to note that the notions of focus and defocus in this study are not drawn on pragmatic information status but rather on semantic alternative sets (Rooth 1992). See §2 for discussion.

2. SFPs are generally present in Cantonese RD (e.g., found in 89.6% of RD cases in Liang 2002's corpus, 94% in Luke 2004's corpus), but may be readily absent in (Mandarin) Mandarin RD (Lu 1980; Yip 2020).

3. Unless specified otherwise, data without sources are constructed by the authors, who are native speakers of Cantonese and Mandarin respectively. The judgment, except for those cited from the literature, is confirmed by 8 other Cantonese consultants (mostly from Hong Kong) and 9 other Mainland Mandarin consultants (mostly from Northern China). Mainland and Taiwan Mandarin are known to differ in prosody (Xu, Chen, and Wang 2012), as well as in RD (Yip 2025a), and

- (6) Baseline [ZP YP SFP]
 [Mingzai wui heoi Meigwok aa3]. [Cantonese]
 [Xiaoming hui qu Meiguo a]. [Mandarin]
 Ming will go US SFP
 ‘Ming will go to the US.’
- (7) Gapped RD [_ YP SFP] ZP
 [_ wui heoi Meigwok aa3] **Mingzai.** [Cantonese]
 [_ hui qu Meiguo a] **Xiaoming.** [Mandarin]
 will go US SFP Ming
 Lit.: ‘Will go to the US, Ming.’ (i.e., ‘Ming will go to the US.’)
- (8) Gapless RD [ZP_i/XP_i YP SFP] ZP_i
 [Mingzai_i/keoi_i wui heoi Meigwok aa3] **Mingzai_i.** [Cantonese]
 [Xiaoming_i/ta_i hui qu Meiguo a] **Xiaoming_i.** [Mandarin]
 Ming/3sg will go US SFP Ming
 Lit.: ‘Ming/he will go to the US, Ming.’ (i.e., ‘Ming will go to the US.’)

It is worth noting that RD, gapped or gapless, is not the same as afterthoughts, and this study only concerns the former. RD and afterthoughts systemically differ in various properties (Frey and Truckenbrodt 2015; Ott and de Vries 2016; Wei and Li 2018; Yip 2020, *i.a.*). More discussions will be given in §2. For now, it suffices to emphasize that RD chunks in Cantonese and Mandarin are always defocused, which sets their RD apart from other constructions that may accommodate focus like afterthoughts, and from other languages where even RD chunks may receive focus (e.g., Japanese and Korean, see Ko 2015; Abe 2019). This provides a crucial ground for testing the role of defocus.

Moreover, Cantonese and Mandarin are also the ideal languages to study prosodic effects of defocus. First, Cantonese does not readily employ pitch prominence for focus and lacks PFC altogether (Wu and Xu 2010; Fung and Mok 2014, *i.a.*), and hence there is presumably no rephrasing from the *presence* of focus. Second, as mentioned above, focus in Mandarin has been shown to have no rephrasing effects despite salient focal prominence and robust PFC (Zhang, Clayards, and Wagner 2024). Therefore, despite the difference in whether focus receives prominence, the *presence* of focus in both languages does not trigger rephrasing, supporting the NO-REPHRASING view. In other words, if we find any (focus-related) syntax-prosody mismatches in RD, one cannot simply attribute them to some kind of post-focal reduction triggered by focus, but rather needs to consider the role of defocus. The effects of focus and defocus can thus be separated.

we focus on the Mainland variety in this paper. Cantonese examples are transcribed using *Jyutping* (the Linguistic Society of Hong Kong Cantonese Romanization Scheme in 1993), and Mandarin examples are transcribed using *Hanyu Pinyin*. Tones are represented for SFPs in Cantonese for disambiguation, otherwise only when necessary (as in §4–5). Tone marks for Cantonese: 1 = high level [55], 2 = high rising [35], 3 = mid-level [33], 4 = low falling [21], 5 = low rising [13], 6 = low level [22]. Tone marks for Mandarin: 1 = high level [55], 2 = high rising [35], 3 = low falling [21] (or falling-rising [214]), 4 = high falling [51]. We follow the Leipzig Glossing Rules with the following exceptions: EXP = experiential aspect; CL = classifier; SFP = sentence-final particles.

1.3 Preview of the proposal

Our research questions are three-fold:

- (9) a. Is there a syntax-prosody mismatch in RD in Cantonese and Mandarin?
- b. If yes, does defocus play a role in triggering the mismatch?
- c. Do the findings support a dissociation of the roles of defocus and focus in prosody?

To answer the research questions, we adopt a rigorous interface approach. Each of our claims is drawn on evidence from different modules of grammar (semantics, syntax, phonology, and phonetics), with explicit assumptions on their mapping.

First, we argue empirically that there is a syntax-prosody mismatch in RD. We will show that RD, while having a *biclausal* structure syntactically (i.e., two CPs), consists of only *one intonational phrase* (henceforth ι) prosodically. In other words, the RD chunk is syntactically a clause but does not map onto an ι , showing syntax-prosody mismatches. Prosodic evidence will be drawn from both domain-sensitive phonological processes and phonetic cues of prosodic boundaries. The latter are examined by acoustic experiments.

Second, we propose that *defocus* is the source of mismatch. We suggest that defocus serves as a mirror image of focus: while focus (often) receives prominence and is mapped onto a prosodic head, defocus cannot receive any prominence and cannot be a prosodic head. Given that RD chunks are always defocused, an illegitimate *headless* ι would be formed if RD chunks were mapped onto an independent ι . Hence, they are rephrased with the preceding main chunks as one ι , leading to syntax-prosody mismatches. This view DEFOCUS REPHRASING view is given in (10).

- (10) The DEFOCUS REPHRASING view: defocus $\rightarrow$ non-head

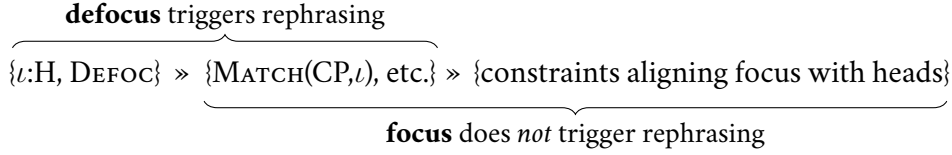
Defocus, unlike focus, cannot receive head prominence and leads to prosodic rephrasing if it is at a certain prosodic domain's edge which would be otherwise aligned with the head.

The core proposal is couched within the Match Theory (Selkirk 2011) with Optimality-Theoretic constraints on the syntax-prosody mapping. There are three core constraints: (i) a mapping constraint $\text{MATCH}(\text{CP}, \iota)$, which maps CPs onto ι (after Selkirk 2011); (ii) a headedness constraint $\iota:\text{HEAD}$ requiring ι to have a prosodic head (*cf.* Selkirk 1996; Ito and Mester 2009); (iii) a *novel* constraint DEFOCUS which prohibits defocus from receiving head prominence (vs. Truckenbrodt 1995's Focus constraint). Importantly, $\iota:\text{HEAD}$ and DEFOCUS are ranked higher than $\text{MATCH}(\text{CP}, \iota)$, as in (11), enabling mismatches for avoiding headless ι .

- (11) $\{\iota:\text{HEAD}, \text{DEFOCUS}\} \gg \text{MATCH}(\text{CP}, \iota)$

Third, this proposal sheds light on the role of (de)focus by representing a weakened version of the REPHRASING view while maintaining the insights from the NO-REPHRASING view. Concretely, while focus may not necessarily be a prosodic head and trigger rephrasing, defocus may trigger rephrasing in the very same language. In other words, there is a *dissociation* of the effects of defocus from those of focus. This maintains a minimal role of focus in syntax-prosody mapping: it is the lack of focus that plays a role, rather than the presence of focus, at least in the case of Cantonese and Mandarin. Represented in OT-constraint ranking, the defocus-related constraints are ranked higher than focus-related constraints such that only defocus but not focus triggers rephrasing, as in (12). This view also implicates that focal prominence is distinct from head prominence, supporting the idea that different prosodic representations should be kept separate (Ishihara 2011, 2016; Wagner and McAuliffe 2019).

(12) Defocus vs. focus under the DEFOCUS REPHRASING view



The rest of this paper contains seven sections. §2 examines the resistance of focus interpretation in RD and argues for the structural notion defocus. §3 demonstrates that both gapped and gapless RD have biclausal structure with typological and language-internal arguments. Turning to prosody, §4 and §5 shows that RD consists of only one intonational phrase, with evidence from phonological processes and acoustic experiments. RD thus displays a syntax-prosody mismatch (two CPs, one ι). §6 proposes an Optimality-Theoretic analysis of how defocus gives rise to the mismatch in RD, and how exactly the RD chunk is prosodically integrated into the main chunk. §7 discusses the implications of the DEFOCUS REPHRASING view on syntax-prosody mapping, concerning both dissociation of focus and defocus and the realization of prosodic (non-)prominence. §8 concludes the paper with prospects on cross-linguistic variations in syntax-prosody mapping.

2 Defocus in right dislocation

In this section, we examine the role of focus in RD. Specifically, we show that RD chunks in both gapped and gapless RD variants systematically resist (alternative-based) focus interpretation, and argue that they should be analyzed as instances of *defocus*, which is a semantic notion with structural reflex cross-linguistically (Zubizarreta 1998; Kallulli 2000; Molnárfi 2002; Zeller 2008; T. T.-M. Lee 2017, 2020).

It is well-documented that RD chunks in Cantonese and Mandarin are defocused (Cheung 2009; T. T.-M. Lee 2017, 2020, *i.a.*), but the exact characterization is subject to debate (see, e.g., Cheung 2009 for information focus and Wei and Li 2018 for contrastive focus), and is often not semantically defined. We follow Rooth (1992) and Krifka (2008) and conceive of “focus” as triggering alternatives in semantics, given in (13). Examples include (i) contrastive focus, (ii) *wh*-question-answer pairs, and (iii) focus particles (e.g., ‘only’, ‘even’, and copula clefts) with their associates.⁴ Defocus is a systematic ban on alternatives, beyond just a ban on contrastive focus. Importantly, defocus is not the same as discourse givenness (*pace* Cheung 2009:203), as will be discussed at the end of this section. Below, we provide a comprehensive examination on (i)–(iii) in RD.

(13) The definition of focus

A focus is an element that triggers a set of alternatives.

First, both gapped and gapless RD resist contrastive stress in RD chunks (Cheung 2015; T. T.-M. Lee 2017, 2023), as illustrated in (14) for Cantonese. Note that the pattern in gapless cases does not change regardless of whether the main chunk has the same contrastive stress.

4. For instance, in *Only Alex_F came*, the focused element *Alex* denotes an entity as its ordinary semantic value, and a set of alternative entities as its focus semantic value ($\llbracket \text{Alex} \rrbracket^o = a$, $\llbracket \text{Alex} \rrbracket^f = \{a, b, c, \dots\}$).

(14) Contrastive focus in Cantonese RD (with stress)

- a. [Zoengsaam __ maai-zo gaa sance aa3] {***CAMJAT_F**/ ^{OK}camjat}.
 Zoengsaam buy-PFV CL new.car SFP yesterday yesterday
 Lit.: 'Zoengsaam bought a new car, *YESTERDAY/yesterday.'
 (gapped RD, adapted from T. T.-M. Lee 2017:68)
- b. [Keoi wui heoi jamngokwui gaa3] {***KEOI_F** wui/ *keoi **WUI_F**/ ***KEOI WUI_F**/
 3sg will go concert SFP 3sg will 3sg will 3sg will
^{OK}keoi wui}.
 3sg will
 Lit.: '(S)he will go to the concert, *(S)HE will/*(s)he WILL/*(S)HE WILL/(s)he will.'
 (gapless RD, adapted from Cheung 2015:261)

Mandarin behaves the same way as Cantonese in disallowing contrastive stress in gapped and gapless RD chunks, as in (15).

(15) Contrastive focus in Mandarin RD (with stress)

- a. [__ bu qu Meiguo a] {***ZHANGSAN_F**/ ^{OK}Zhangsan}. (gapped RD)
 not go US SFP Zhangsan Zhangsan
 Lit.: 'will not go to the US, *ZHANGSAN/Zhangsan.'
- b. [Zhangsan bu qu Meiguo a] {***ZHANGSAN_F**/ ^{OK}Zhangsan}. (gapless RD)
 Zhangsan not go US SFP Zhangsan Zhangsan
 Lit.: 'Zhangsan will not go to the US, *ZHANGSAN/Zhangsan.'

Even without stress, contrastive focus is still not possible in RD chunks. This can only be tested in Cantonese where focal stress is not obligatory (see §7). As reported by T. T.-M. Lee (2020, 2023), the RD chunks cannot accommodate a contrastive reading in (16a), as opposed to (16b) with the contrastive focus in the main chunks.

(16) Contrastive focus in Cantonese RD (without stress)

- a. *[Ngo heoi-gwo __ aa3] **Hoenggong_F**. [Daan mou heoi-gwo __ aa3] **Oumum_F**.
 1sg go-EXP SFP Hong Kong but not go-EXP SFP Macau
 Int.: 'I have been to *Hong Kong*. But (I) have not been to *Macau*.'
 (T. T.-M. Lee 2020:141, notation # changed to * under the intended reading)
- b. [__ heoi-gwo **Hoenggong_F** aa3] ngo. [Daan __ mou heoi-gwo **Oumum_F** aa3] ngo.
 go-EXP Hong Kong SFP 1sg but not go-EXP Macau SFP 1sg
 'I have been to *Hong Kong*. But I have not been to *Macau*.'

Second, RD chunks also cannot be *wh*-words, or answers to *wh*-questions, both of which trigger alternatives (following Rooth 1992; Beck 2006; Erlewine 2025). The Cantonese facts are noted by Cheung (2009) and T. T.-M. Lee (2017, 2020) for gapped RD, and we extend this observation to gapless RD, as illustrated in (17) for *wh*-words and (18) for answers to *wh*-questions.⁵

5. We noticed that Chan (2016, 16) reported *wh* in gapless RD as acceptable in Cantonese, but his judgment could not be replicated for our 8 consultants.

(17) Wh-words in Cantonese RD

- a. Camjat [bin jat-go hoksaang]_F lai-zo ne1? (canonical)
 yesterday which one-CL student come-PFV SFP
 ‘Which student came yesterday?’
- b. *[Camjat {__ / [bin jat-go hoksaang]_F} lai-zo ne1] [**bin jat-go hoksaang**]_F? (RD)
 yesterday which one-CL student come-PFV SFP which one-CL student
 Lit.: ‘(Which student) came yesterday, which student?’

(18) Answers to wh-questions in Cantonese RD

- Q. Which student came yesterday? (=17a)
- A1. Camjat Gaajan_F lai-zo aa3. (canonical)
 yesterday Ka-Yan come-PFV SFP
 ‘Ka-Yan came yesterday.’
- A2. *[Camjat {__ / Gaajan_F} lai-zo aa3] **Gaajan_F**. (RD)
 yesterday Ka-Yan come-PFV SFP Ka-Yan
 Lit.: ‘(Ka-Yan) came yesterday, Ka-Yan.’

In Mandarin, both gapped and gapless RD constructions are incompatible with the RD chunks being *wh*-words as in (19), or answers to *wh*-questions as in (20).

(19) Wh-words in Mandarin RD

- *[Mingtian {__ / shei_F} hui qu a] **shei_F**? (RD)
 tomorrow who will go SFP who
 Lit.: ‘(Who) will go tomorrow, who?’ (Int.: ‘Who will go tomorrow?’)

(20) Answers to wh-questions in Mandarin RD

- Q. Who will go tomorrow?
- A. *[Mingtian {__ / Lisi_F} hui qu a] **Lisi_F**. (RD)
 tomorrow Lisi will go SFP Lisi
 Lit.: ‘(Lisi) will go tomorrow, Lisi.’

Third, RD chunks cannot contain focus particles with their associates. For example, exclusive focus ‘only’ cannot occur in both gapped and gapless RD chunks in Cantonese (T. T.-M. Lee 2023), as shown in (21).

(21) ‘Only’ focus in Cantonese RD

- a. Zinghai ngo_F maai-zo ni-bun syu zaa3. (baseline)
 only 1SG buy-PFV this-CL book SFP
 ‘Only I bought this book.’
- b. ??[{__ / zinghai ngo_F} maai-zo ni-bun syu zaa3] **zinghai ngo_F**. (RD)
 only 1SG buy-PFV this-CL book SFP only 1SG
 Lit.: ‘bought this book, only I.’ (i.e. ‘Only I bought this book.’) (T. T.-M. Lee 2023:467)

In Mandarin, again, both gapped and gapless RD chunks with ‘only’ focus are unacceptable as in (22).

(22) ‘Only’ focus in Mandarin RD

- *[{ __ / zhiyou wo_F } maai-le na-bun shu eryi] **zhiyou** wo_F. (RD)
 only 1SG buy-PFV this-CL book SFP only 1SG
 Lit.: ‘(Only I) bought this book, only I.’ (i.e. ‘Only I bought this book.’)

As a novel observation by us, focus particles (operators) behave differently from focus associates (focused elements) in RD. While focus particles alone may occur in the RD chunks, as illustrated for Cantonese in (23a) and (23c) (see also Cheung 2009, 2015; Yip 2025a), focus associates by themselves cannot appear in RD chunks as in (23b) and (23d). This is consistent with the definition of focus in (13): it is the focus associate that triggers the alternatives (i.e., {this book, that book, ...}), but not the focus particle ‘only’, which only operates on the alternative sets (Rooth 1992). Hence, the focus associates but not the focus particles ($\neq$ focus *per se*) are banned in RD chunks.

(23) ‘Only’ particles vs. ‘only’ focus associates in Cantonese RD

- a. [Aaming __ maai-zo [ni-bun syu]_F aa3] **zinghai**. (particle, gapped RD)
 Ming buy-PFV this-CL book SFP only
 Lit.: ‘Ming bought *this book*, only.’ (i.e. ‘Ming only bought *this book*.’)
- b. *[Aaming **zinghai** maai-zo __ aa3] [**ni-bun syu**]_F. (Obj. focus, gapped RD)
 Ming only buy-PFV SFP this-CL book
 Lit.: ‘Ming only bought, *this book*.’
- c. [Zinghai Aaming_F wui heoi zaa3] **zinghai**_F. (particle, gapless RD)
 only Ming will go SFP only
 Lit.: ‘Only *Ming* will go, only.’ (i.e. ‘Only *Ming* will go.’)
- d. *[**Zinghai** Aaming_F wui heoi zaa3] **Aaming**_F. (Subj. focus, gapless RD)
 only Ming will go SFP Ming
 Lit.: ‘Only *Ming* will go, *Ming*.’

The same patterns are found in Mandarin with adverbial *zhishi* ‘only’ and adnominal *zhiyou* ‘only’ (see Sun 2021 for the adverbial-adnominal distinction), as in (24).⁶

(24) ‘Only’ particles vs. ‘only’ focus associates in Mandarin RD

- a. [Zhangsan { __ / zhishi } jie-le [na-ben xiaoshuo]_F ba] **zhishi**. (adverbial prt.)
 Zhangsan only borrow-PFV that-CL novel SFP only
 Lit.: ‘Zhangsan (only) bought *this novel*, only.’ (i.e., ‘ZS only bought *this novel*.’)
- b. *[Zhangsan **zhishi** jie-le { __ / [na-ben xiaoshuo]_F } ba] [**na-ben xiaoshuo**]_F.
 Zhangsan only borrow-PFV that-CL novel SFP that-CL novel
 Lit.: ‘Zhangsan only bought (*this novel*), *this novel*.’ (Obj. focus)
- c. Context: Only who came?
 [Keneng { __ / zhiyou } Lisi_F lai-le ba] **zhiyou**_F. (adnominal prt.)
 perhaps only Lisi come-PFV SFP only
 Lit. ‘Perhaps (only) *Lisi* came, only.’ (i.e., ‘Perhaps only *Lisi* came.’)

6. Monosyllabic *zhi* ‘only’ cannot be right-dislocated for reasons unknown to us.

- d. *[Keneng **zhiiyou** {__ / Lisi} lai-le ba] **Lisi**_F. (Subj. focus)
 perhaps only Lisi come-PFV SFP Lisi
 Lit.: ‘Perhaps only (Lisi) came, *Lisi*.’

Apart from ‘only’, other focus particles with their associates are also disallowed in RD chunks. Cleft focus formed by copula (*hai* in Cantonese, *shi* in Mandarin) cannot occur in both gapped and gapless RD chunks (Chiang 2017, 2022; T. T.-M. Lee 2023). RD of the focus associates alone is not possible either. For space reasons, we give one example each for the two languages in (25–26).

(25) Copula cleft focus in Cantonese RD

- *[Mingzai {__ / hai tingjat_F} lai lo1] **hai tingjat**_F. (cop+focus)
 Ming COP tomorrow come SFP COP tomorrow
 Int.: ‘It is tomorrow that Ming will come.’

(26) Copula cleft focus in Mandarin RD

- *[Lisi **shi** {__ / mingtian_F} yao lai a] **mingtian**_F. (focus)
 Lisi cop tomorrow will come SFP tomorrow
 Int.: ‘It is tomorrow that Lisi will come.’

‘Even’ focus patterns in the same way. Both gapped RD and gapless RD disallow ‘even’ focus in RD chunks, marked by *lin...dou* in Cantonese and *lian...ye* in Mandarin (T. T.-M. Lee 2020, 2023; Yip 2025a), as in (27). Similarly, focus associates alone are disallowed in the RD chunks as in (28).

(27) ‘Even’ focus in Cantonese RD

- *[Ngo {__ / lin bouzi_F} dou m-wui tai gaa3] **lin bouzi**_F (**dou**). (prt.+focus)
 1sg even newspaper also not-will read SFP even newspaper also
 Int.: ‘I will not read even *newspapers*.’ (adapted from T. T.-M. Lee 2023:468)

(28) ‘Even’ focus in Mandarin RD

- *[Lousi **lin** {__ / GB_F} **ye** bu jiao a] **GB**_F! (focus)
 teacher even GB also not teach SFP GB
 Lit.: ‘The teacher, even *GB*, doesn’t teach (it), *GB*!’

Taking stock, the tests of contrastive focus, *wh*-question-answer pairs, and associates of focus particles all show that RD does not allow focus, as summarized in Table 1. We conclude that RD chunks are always defocused.

Type	Focus	Cantonese		Mandarin	
		gapped RD	gapless RD	gapped RD	gapless RD
Contrastive	With stress	✗	✗	✗	✗
	Without stress	✗	✗	N/A	N/A
Q-A	<i>Wh</i> -words	✗	✗	✗	✗
	Ans to <i>wh</i>	✗	✗	✗	✗
Associates of focus particles	Copula cleft	✗	✗	✗	✗
	‘Only’	✗	✗	✗	✗
	‘Even’	✗	✗	✗	✗

Table 1: The ban on focus in RD chunks

(29) Q. Why were you so mad yesterday? [Mandarin]
 A. [{__ / Wo gege} dapo-le wo-de diezi a] **wo gege**.
 1SG older.brother break-PFV 1SG-DE plate SFP 1SG older.brother
 Lit.: '(My brother) broke my plate, my brother.'

(30) Q. What happened to Ming's leg? [Cantonese]
 A. [Keoi zek goek { __ / bei mau-go jan} daatun-zo aa3] **bei mau-go jan**.
 3SG CL foot PASS some-CL person hit.break-PFV SFP PASS some-CL person
 Lit.: 'His leg is beaten to broken (by someone), by someone.'

(31) RD is sensitive to syntactic islands

T. T.-M. Lee (2017, 2020) propose that RD chunks undergo movement to a defocus projection (DeFocP) triggered by a $[-\text{Foc}]$ feature (vs. $[\text{+Foc}]$), deriving the defocus interpretation. He also of-

fers arguments for the syntactic activeness of the feature from (non-)intervention effects (T. T.-M. Lee 2022, 2024). We follow him and suggest that RD chunks in Cantonese and Mandarin occupy a designated syntactic slot, DeFocP. Similar defocus projections have been proposed across languages with morphosyntactic reflexes, such as Spanish and Italian for P-movement (Zubizarreta 1998), West Germanic for scrambling (Molnárfi 2002), Albanian and Greek for object clitic doubling (Kallulli 2000), and Bantu for subject/anti-focus markers (Zeller 2008), among others. Notably, both strategies of movement and morphological markers are found, the latter mirroring focus morphology (e.g., focus concord markers). Hence, defocus is a cross-linguistically attested syntactic category, which may either attract movement or be realized as agreement-like markers. Semantically, $[-\text{Foc}]$ is interpreted as the lack of alternatives.⁷

Second, RD is different from afterthoughts (Ott and de Vries 2016; Wei and Li 2018; Yip 2020, *i.a.*), in terms of information structure, syntactic properties, as well as prosodic properties (see §6.1). One example of afterthoughts is given in (32), where the afterthought element ‘woolen’ supplements information missing in the first sentence (i.e., modifying the object). Syntactically, the afterthought chunk may have an additional SFP (Wei and Li 2018), which is impossible in RD. Of interest here is that afterthoughts may accommodate focus, unlike RD, as illustrated by the contrastive focus in (33).

- (32) [Wo mai-le ding maozi], nizi-de (**ne**). (afterthought, $\neq$ RD) [Mandarin]
 I buy-PFV CL hat woolen SFP.
 Lit.: ‘I bought a hat, woolen.’ (adapted from Wei and Li 2018:274)
- (33) *Context: A asked: “Did the boss tell you how many kilograms to buy for each type of bananas?” B answered: “Yes. 2 kgs each for the golden banana and the silver banana.” A knew that only silver bananas require 2 kgs, and responded: “Eh? That’s not right. ...”*
 ... [Yinghai SAN gongjin a], **JIN_F** **xiangjiao**. (afterthought, $\neq$ RD) [Mandarin]
 should three kg SFP golden banana
 ‘It should be 3kgs, the GOLDEN banana.’ (adapted from Wei and Li 2017)

3 Syntax: RD is biclausal

The section argues that RD in Cantonese and Mandarin has an underlying biclausal structure, i.e., with two CPs. We first briefly review previous syntactic approaches to RD. There are two approaches with different assumed number of underlying clauses. The monoclausal approach assumes that there is only one clause, and the two chunks are derivationally related by movement (Cheung 1997, 2009; Law 2003; T. T.-M. Lee 2017, 2021, *i.a.*). As illustrated in (34), the RD chunk moves to a defocus projection, followed by remnant movement of the main chunk above the SFP, which is assumed to be head-initial (e.g., Simpson and Wu 2002; Pan 2022). The gapless variant involves pronouncing the XP trace/copy (T. T.-M. Lee 2021).

- (34) The monoclausal approach to RD (to be rejected)
- a. $[_{\text{CP}} \text{SFP } [_{\text{DeFocP}} \text{XP}_{\text{RD}} \text{DeFoc-}\emptyset \text{ } [_{\text{main}} t_{\text{XP}} \text{YP}]]]$
- b. $[_{\text{CP}} \text{ } [_{\text{main}} \{t_{\text{XP}} / <\text{XP}>\} \text{YP}] \text{SFP } [_{\text{DeFocP}} \text{XP}_{\text{RD}} \text{DeFoc-}\emptyset t_{\text{main}}]]$ (XP-)YP-SFP-XP

7. Formally, we suggest that $\text{XP}_{[-\text{Foc}]}$ presupposes that the focus semantic value is the singleton set containing the ordinary semantic value: $\llbracket \text{XP} \rrbracket^f = \{\llbracket \text{XP} \rrbracket^o\}$, similar to the resetting operator $\sim$ (Rooth 1992), except that it is presuppositional here.

The biclausal approach, in contrast, posits two underlying “parallel” clauses that are juxtaposed or coordinated (Cheung 2015; Tang 2015a, 2018; Yip 2025a, *i.a.*; see also Ott and de Vries 2016). In CP2, the RD element undergoes the same defocus movement, with the remnant part subsequently sluiced/deleted, as in (35). It is compatible with both head-initial and head-final SFPs. The gapped and gapless variants only differ in whether empty categories like subject *pro* (notated as *e*) are used in CP1, the main chunk (Chan 2016; Dong 2018; Yip 2025a).

(35) The biclausal approach to RD (to be argued for)

- a. $[_{CP1-main} \{e_i / XP_i\} YP SFP] [_{CP2-RD} [_{DeFocP} \mathbf{XP}_i DeFoc-\emptyset [t_{XP} YP SFP]]]$
- b. $[_{CP1-main} \{e_i / XP_i\} YP SFP] [_{CP2-RD} [_{DeFocP} \mathbf{XP}_i DeFoc-\emptyset [\cancel{t_{XP}} \cancel{YP SFP}]]]$ (XP-)YP-SFP-XP

In the following, we provide a typological argument for the biclausal approach in §3.1, and reproduce and enrich two language-internal arguments from the literature in §3.2. Our main goal is to justify the existence of two CPs in RD, rather than to adjudicate the minute details of various proposals (see Yip 2025a for a recent critical overview).

3.1 A typological argument

First, cross-linguistically, the inventory of empty categories strongly correlates with the availability of gapped RD. Roughly speaking, only null argument languages allow gapped RD of arguments. These include East Asian languages like Chinese (see the citations above), Japanese (Kuno 1978; Abe 1999; Tanaka 2001, *i.a.*; see Takita 2011 for overview), Korean (Park and Kim 2009; Chung 2009; Yim 2013, *i.a.*; see Ko 2022 for overview), as well as Alasha Mongolian (T. T.-M. Lee 2023). Some examples are illustrated in (36–39) (*e*=empty category).

(36) Mandarin: ✓ Empty subjects, ✓ Gapped subject RD

Context: Lisi asks you whether Zhangsan will come. You answer:

- a. e_S hui lai a. b. [e_S hui lai a] **Zhangsan**.
 will come SFP will come SFP Zhangsan
 ‘(Zhangsan) will come.’ ‘(He) will come, Zhangsan.’

(37) Cantonese: ✓ Empty objects, ✓ Gapped object RD

Context: Tommy is showing off his new MacBook. You say:

- a. Ngo dou jau e_O laa1. b. [Ngo dou jau e_O laa1] **mek1 gei1**.
 1SG also have SFP 1SG also have SFP Mac computer
 ‘I also have (a Mac).’ ‘I also have (one), a Mac.’

(38) Korean: ✓ Empty subjects, ✓ Gapped subject RD

((b) from Ko 2022:359, context added)

Context: You’re discussing about Cheli.

- a. e_S ecey Yenghi-lul manna-ss-e. b. [e_S Ecey Yenghi-lul manna-ss-e] Cheli-ka.
 ytd. Y.-ACC meet-PST-DECL ytd. Y.-ACC meet-PST-DECL C.-NOM
 ‘(Cheli) met Yenghi yesterday.’ ‘(He) met Yenghi yesterday, Cheli.’

(39) Japanese: ✓ Empty object, ✓ Gapped object RD

((b) from Tanaka 2001:551, context added)

Context: Someone asks who read LGB, and you answer:

- a. John-ga e_o yonda yo.
John-NOM read SFP
'John read (it).'
- b. [John-ga e_o yonda yo] **LGB-o**.
John-NOM read SFP LGB-ACC
'John read (it), LGB.'

On the other hand, languages that disallow subject and object drop, such as Dutch, German, and English, do not allow subject and object RD to be gapped. A resumptive pronoun is obligatory in the main chunk (Ott and de Vries 2016), as illustrated in (40–41).

(40) English: ✗ Empty subjects, ✗ Gapped subject RD

- a. Apparently ***(he)** is very nice. b. [Apparently ***(he)** is very nice], **John**.

(41) Dutch: ✗ Empty objects, ✗ Gapped object RD (adapted from Ott and de Vries 2016:656)

- a. Ik heb ***(m)** gezien.
I have him seen
'I saw him.'
- b. [Ik heb ***(m)** gezien], **die man**.
I have him seen that man
'I saw him, that man.'

In the monoclausal approach, gaps are analyzed as movement traces, as in (42a). In contrast, they are proposed to be empty categories in the biclausal approach, as in (42b).

- (42) a. $[CP [TP [t_{XP}] YP] [SFP [XP_{RD} \dots t_{TP}]]]$ Monoclausal: *YP-SFP-XP*
b. $[CP_1 [e_i] YP SFP] [CP_2 XP_{RD_i} [\dots t_{XP} YP SFP]]]$ Biclausal: *YP-SFP-XP*

Given that subjects and objects can move and leave traces in non-argument-drop languages (e.g., *wh*-movement of arguments, subject raising, passivization of objects, etc.), the obligatoriness of resumptive pronouns in RD in the above-mentioned Germanic languages is unexpected under the monoclausal approach. On the other hand, the contrast between these two sets of languages falls out from the biclausal approach—only languages with null arguments allow argumental gaps in RD.

To further strengthen this correlation, consider a major set of Romance languages that allow subject drop but not object drop, such as Catalan, Italian, European Portuguese, Spanish, etc. (excluding French). In these languages, only gapped subject RD is allowed, but not gapped object RD (see Zubizarreta 1998; Fernández-Sánchez 2017 and references therein), as shown in (43–44). RD of objects requires clitics in the main chunks.⁸

(43) Catalan: ✓ Empty subjects, ✓ Gapped subject RD (Fernández-Sánchez 2017:91)

- a. e_s és molt maco.
is very nice
'He is very nice.'
- b. [e_s és molt maco], **en Joan**.
is very nice the Joan
'He is very nice, Joan.'

(44) Spanish: ✗ Empty objects, ✗ Gapped object RD (adapted from Zubizarreta 1998:154)

- a. ***(La)** lavó mamá.
it wash-PST.3SG mom
'Mom washed ***(it)**.'
- b. [***(La)** lavó mamá], la mamadera.
it wash-PST.3SG mom the milk.bottle
'Mom washed it, the bottle.'

8. Zubizarreta notes that the string in (44b) is grammatical only if: (i) the subject bears contrastive (instead of nuclear) stress, (ii) the object is deaccented, and (iii) there is no prosodic break before. She argues this to be a case with in-situ objects, and concludes that clitics in the main chunk are still obligatory in object RD. See also Cecchetto (1999) and Cardinaletti (2002) for relevant discussion in Italian.

Again, this correlation can only be captured under the biclausal approach, where gapped object RD is unavailable due to the lack of object drop, whereas gapped subject RD is possible with subject drop.

Another interesting correlation is whether empty verbs and gapped verb RD are allowed. In Cantonese and Mandarin, copulas and (non-copular) verbs can be null in non-elliptical contexts (Tang 1999, 2001b, 2001a). Strikingly, they can also be gapped in RD in both languages, as in (45–46). Note that empty verbs require licensing by a focused object (Tang 1999). The very same licensing condition has also been observed for gapped verb RD (T. T.-M. Lee 2021:123–125).⁹

(45) Cantonese: ✓ Empty copula, ✓ Gapped copula RD

- | | |
|---|--|
| <p>a. Gamjat <i>e_{COP}</i> singkeijat aa3.
 today Sunday SFP
 ‘Today is Sunday.’</p> | <p>b. [Gamjat <i>e_{COP}</i> singkeijat aa3] hai.
 today Sunday SFP COP
 ‘Today is Sunday.’</p> |
|---|--|

(46) Mandarin: ✓ Empty verb, ✓ Gapped verb RD

- a. Zhangsan *e_V* SAN-ge pingguo, Lisi *e_V* SI-ge juzi.
Zhangsan three-CL apple Lisi four-CL orange
‘Zhangsan (bought, ate, etc.) *three* apples, and Lisi *four* oranges.’ (Tang 2001b:205)
- b. [Zhangsan *e_V* SAN-ge pingguo ma] {**yao/ you/ mai-le...**} ?
Zhangsan three-CL apple SFP want have buy-PFV
‘Does/did Zhangsan {want/ have/ buy} *three* apples?’

On the other hand, empty verbs—in non-elliptical contexts—are banned in the above-mentioned Germanic and Romance languages, and even in Japanese. Correspondingly, they also lack gapped copula/verb RD, as illustrated in (47–49).^{10, 11}

(47) English: ✗ Empty copula, ✗ Gapped copula RD

- | | |
|---|---|
| <p>a. John *(is) real smart.</p> | <p>b. *John real smart, is.</p> |
|---|---|
- c. cf. He’s real smart, **John is**. (Kayne 1994:78)

(48) Spanish: ✗ Empty verb, ✗ Gapped verb RD

Context: Who saw Guille yesterday?

- | | |
|--|--|
| <p>a. Juan *(vio) a Guille ayer.
 Juan see-PST.3SG ACC Guille yesterday
 ‘Juan saw Guille yesterday.’</p> | <p>b. *Juan <i>e_V</i> a Guille ayer, vio.
 Juan ACC Guille yesterday see-PST.3SG
 Lit.: ‘Juan Guille yesterday, saw.’
 (Alessandra Pintado-Urbanc, María Teresa Borneo, p.c.)</p> |
|--|--|

9. As reported by Yip (2025a), unlike Mainland Mandarin, Taiwan Mandarin does not allow gapped verb RD, although empty verbs are available. Yet, even gapless RD with the verbs copied twice is not possible in Taiwan Mandarin. Thus, this is not a counter-example to the correlation under discussion.

10. We do not have access to Catalan and Dutch speakers to verify this point, but the same ban on gapped verb RD is found in German (Anton Hampe, p.c.) and Korean (Youngdong Cho, Yumin Ji, p.c.).

11. “(Elliptical) gapping” as in *John talked about Bill and Mary talked about Susan*, whose availability relies on a linguistic antecedent (unlike empty categories), should not be considered.

(49) Japanese: ✗ Empty verb, ✗ Gapped verb RD

Context: *Who did you give the book to?*

- a. Watashi-wa Tanaka-ni hon-o *(**age-ta**) yo.
1SG-TOP Tanaka-DAT book-ACC give-PST SFP
'I gave Tanaka a book.'
- b. *[Watashi-wa Tanaka-ni hon-o *e_V* yo] **age-ta**.
1SG-TOP Tanaka-DAT book-ACC SFP give-PST
Lit.: 'I Tanaka a book, gave.'

(Yusuke Yagi p.c.)

These patterns are nicely captured by the biclausal approach, where verbal gaps in RD are base-generated empty verbs. They are independently available in Chinese (see Tang 2001b for justification) but not in the other languages discussed here. While the monoclausal approach could resort to the unavailability of verb movement, this attempted explanation faces immediate difficulties since English copula can move as in *is John real smart?*, but (47b) is still ungrammatical (as opposed to (47c) with the copula in the main chunk). A further challenge comes from the fact that Chinese, unlike (most) Germanic languages (e.g., Hein 2021), requires verb doubling in predicate clefting (Cheng and Vicente 2013; T. T.-M. Lee 2021, 2024), as in (50). That is, verbal gaps are independently unavailable in Chinese, but gapped verb RD is still attested, suggesting that the gaps in RD should not be attributed to movement, but to empty categories.¹²

- (50) a. **Soeng**, Aaming hai *(**soeng**) sik jyu ge2. [Cantonese]
want Ming COP want eat fish SFP
'As for (whether he) wants, Aaming wants to eat fish.' (T. T.-M. Lee 2021:110)
- b. **Chi**, wo shi *(**chi-guo**) le, ... [Mandarin]
eat I COP eat-EXP SFP
'As for eating, I have (indeed) eaten, ...' (adapted from Cheng and Vicente 2013:2)

To summarize, whether or not an empty category is attested in a given language correlates with whether or not gapped RD of the same type is available, as in Table 2. This strong typological generalization can only be captured under the biclausal approach to RD.

Languages	Empty category inventory			Gapped right dislocation		
	Subj. drop	Obj. drop	Empty V/cor	Subj. gap	Obj. gap	V/cor gap
Cantonese, Mandarin	✓	✓	✓	✓	✓	✓
Japanese, Korean, ...	✓	✓	✗	✓	✓	✗
Catalan, Spanish, ...	✓	✗	✗	✓	✗	✗
Dutch, German, English, ...	✗	✗	✗	✗	✗	✗

Table 2: Correlation between empty categories and gapped RD

12. Although no gaps are allowed, verb doubling in Chinese is still analyzed as movement with multiple copies pronounced (e.g., Cheng and Vicente 2013). See T. T.-M. Lee (2021) for a Cyclic Linearization account.

3.2 Language-internal arguments

There is also ample Chinese-internal evidence for the biclausal structure in RD, as thoroughly discussed in Yip (2025a). We reproduce and further enrich two such arguments in Chinese: one for gapless RD, and one for gapped RD.¹³

First, in gapless RD, the RD chunk may be distinct from its correlate in the main chunk, dubbed “imperfect copying” by Cheung (2015). Two naturally-occurring examples are given in (51).

(51) Imperfect copying in gapless RD

- a. [Go-tou_i dou hou ging gaa], go-bou hei_i. [Cantonese]
 that-CL also very awesome SFP that-CL movie
 ‘The movie is also awesome.’ (Cheung 2015:272, from corpus)
- b. Wo kan, [zhe_i zhi-bu-zhu le], na-ge_i. [Mandarin]
 1SG see this hold-not-still SFP that-CL
 ‘In my view, this cannot hold still anymore, that one.’
 (Meng 1982:175, from daily conversation)

We note that these cases lack a monoclausal source, since putting the RD chunk back to the main chunk yields ungrammaticality, like the illicit double classifiers in (52a) and illicit double demonstratives in (52b). Under the monoclausal approach where gapless RD is analyzed as multiple copy pronunciation (e.g., T. T.-M. Lee 2021), imperfect copying is unexpected.¹⁴ On the other hand, the biclausal approach naturally predicts examples like (51) since the RD chunks and their correlates are base-generated in different clauses, as shown in (53).

- (52) a. *Go-tou (go-)bou hei dou hou ging gaa. [Cantonese]
 that-CL that-CL movie also very awesome SFP
- b. *Wo kan, zhe na-ge zhi-bu-zhu le. [Mandarin]
 1SG see this that-CL hold-not-still SFP
- (53) a. [CP1 That-CL_{tou_i} is also awesome] [CP2 that-CL_{bou} movie_i [—]]
- b. [CP1 This_i cannot hold still anymore] [CP2 that one_i [—]]

The second argument concerns whether negation can be right-dislocated. Yip (2025a) observes an intriguing contrast between gapped and gapless RD. As in (54), negation can only be copied, but not gapped.

(54) Negation cannot be gapped in RD

- a. [Keoi *(zung mei) heoi-gwo Meigwok gaa3] zung mei. [Cantonese]
 3SG still not.yet go-EXP US SFP still not.yet
 ‘S/he hasn’t been to the US yet.’

13. Other biclausal arguments offered by Yip (2025a) include: (i) asymmetries between the two chunks in reflexive binding, universal *wh*-licensing, ‘even’-focus licensing, and NPI-licensing; (ii) parallel referential potential of subject gaps to *pro*; (iii) SFP cluster ordering; and (iv) apparent focus association into main chunks.

14. One might suggest an alternative with partial Copy Deletion, which may reduce lexical XPs to pronoun copies (e.g., van Urk 2018; see Yip and Ahenkorah 2023; Yip and Yuan 2024 for Cantonese and Mandarin). While this could capture imperfect copying with resumptive pronouns (e.g., *I know him_i, Peter_i*), the classifiers and demonstratives in (51) have distinct SynSem features and cannot be reduced to one another.

- b. [Ta *(**bu hui**) qu Meiguo a] **bu hui**. [Mandarin]
 3SG not will go US SFP not will
 ‘S/he won’t go to the US.’ (Yip 2025a: ex.42–43)

Yip (2025a) argues that the contrast is unexpected under the monoclausal approach since the negation should have been possible to be reconstructed back to the main chunk (see, e.g., Cheung 2009; T. T.-M. Lee 2017 for reconstruction for binding in RD). On the contrary, the biclausal approach assumes that negation does not have a corresponding empty category. In gapped RD, CP1 denotes an affirmative proposition, which contradicts the negative CP2 that it coordinates/juxtaposes with. In gapless RD, both CPs denote negative polarity and are compatible with each other.

- (55) a. # $[_{CP1} \dots (\text{affirmative}) \dots \text{SFP}] [_{CP2} \text{Neg}_i [\dots t_i \dots]]$ (gapped RD: contradiction)
 b. $[_{CP1} \dots \text{Neg} \dots \text{SFP}] [_{CP2} \text{Neg}_i [\dots t_i \dots]]$ (gapless RD: consistency)

We further strengthen this argument with a case of Neg-raising. Luke (2004) observes that (56) lacks a monoclausal source: if the RD chunk is “put back” to the main chunk, a double negation reading arises, which is different from the single negation reading in RD.

- (56) a. [Bu re ma], ni bu juede? b. #Ni bu juede bu re ma? [Mandarin]
 not hot SFP 2SG not think 2SG not think not hot SFP
 ‘Isn’t it hot, don’t you think?’ ‘Don’t you think it’s not hot?’ $\neq$ (a)
 ((a) from Luke 2004:42)

With Neg-raising predicates like *juede* ‘think’, it is the embedded proposition that is negative (e.g., *I don’t think it’s hot* $\rightsquigarrow$ *I think it’s not hot*). The two CPs, under the biclausal approach, are thus consistent with each other and can be juxtaposed to form RD.

- (57) $[_{CP1} \text{Isn’t it hot?}] [_{CP2} \text{Don’t you think it’s hot?}]$

In sum, we have argued that both gapped and gapless RD consist of two underlying CPs. The next section proceeds to the prosody of RD—which, as we will see, mismatches with the syntax.

4 Prosody: RD forms one intonational phrase

This section draws phonological evidence to argue that RD forms one intonational phrase. Coupled with the biclausal syntax of RD, we see a syntax-prosody mismatch in RD, as indicated in (58) by the shading on CP boundaries that have no corresponding ι boundaries in prosody.

- (58) The prosodic mismatch in of RD (only ι shown): Two CPs, one ι

$$\begin{array}{c} \text{[}_{CP1\text{-main}} \text{ (XP) YP SFP] } \text{] } \text{ [}_{CP2\text{-RD}} \text{ [}_{DeFocP} \text{ XP } \text{YP SFP} \text{] } \text{] } \\ \text{(} \hspace{15em} \text{)}_{\iota} \end{array} \quad \begin{array}{l} \text{syntax} \\ \text{prosody} \end{array}$$

Two domain-sensitive phonological phenomena are recruited to test the prosodic phrasing. The first one is boundary tone placement in Cantonese, which applies on the intonational phrase ι level. The second one is third tone (T3) sandhi in Mandarin, which applies on the phonological phrase ϕ level or below. These phenomena will help us to determine whether RD has one ι as a whole, two side-by-side ι , or a (left-headed) recursive ι (where the maximal one is marked as ι_{max}):

- (59) $\left[\text{CP1-main (XP) YP SFP} \right]] \left[\text{CP2-RD} \left[\text{DeFocP XP } \cancel{\text{YP SFP}} \right] \right]$ *two CPs* *syntax*
- a. () $_{\iota}$ *one ι* *prosody*
- b. () $_{\iota_1}$ () $_{\iota_2}$ *two ι* *prosody*
- c. (() $_{\iota}$) $_{\iota_{max}}$ *recursive ι* *prosody*

For instance, if there is only one ι , then any phonological phenomena that pick out the right edge of the ι cannot occur at the end of main chunks, before RD chunks. This is the case for Cantonese boundary tones, as we show below in §4.1. Similarly, any phonological phenomena that are blocked by edges of ι should only be allowed in non- ι -edges, hence expected to occur in-between the two chunks under the one- ι phrasing, which is precisely the case of third tone sandhi in Mandarin, as will be discussed in §4.2. We will also confirm the phonological phrasing by phonetic cues from acoustic experiments in the next section §5. Note that tones will be given for the examples in this section and next for transparency (see fn.3 for the inventories).

4.1 Placement of boundary tones in Cantonese

Cantonese has a rich set of boundary tones, as given in the C-ToBi inventory in Table 3. These boundary tones can only occur at the right edges of intonational phrases, and there are no left boundary tones (Wong, Chan, and Beckman 2005).

No.	Tone types	Descriptions
1	L%	fall from the final lexical tone
2	H%	rise from the final lexical tone
3	H:%	rise from the final lexical tone, with a short plateau at the end
4	HL%	final rise and then fall from the final lexical tone
5	%	phrase-end with no extra tone
6	-%	truncated rise of the final lexical tone
7	%fi ¹⁵	frame-initial boundary used to mark the initial particle <i>mat</i>

Table 3: The inventory of boundary tones in C-ToBi (Wong, Chan, and Beckman 2005:287)

Take the boundary tone H% as an illustration. It realizes as local f0 rising on the last syllable of an ι to indicate a question (Wong, Chan, and Beckman 2005; Xu and Mok 2011; Zhang 2014). In (60), the mid-level tone *gwok3* [k^wɔk³³] is pronounced with a high rising boundary tone as [k^wɔk³⁵].

- (60) The rising boundary tone H% occurs at right edges of ι
(Ming4zai2 wui3 heoi3 Mei5**gwok3**) $_{\iota}$ ^{H%} ? (gwok3 pronounced as [k^wɔk³⁵])
Ming will go US
‘Ming will go to the US?’

The boundary tone H% is in complementary distribution with question particles (Zhang 2014), such as *aa4* (with low falling tone, [a²¹]), as in (61). Here, both *gwok3* and *aa4* retain their lexical tone as [k^wɔk³³] and [a²¹] respectively. H% and *aa4* share similar semantic functions: they mark a yes-no question with positive evidential bias (Yip 2025b), in a way similar to English rising declarative. Tang

15. %fi is used for a segmental sentence-initial particle *mat* in questions (Tang 2008; Cheng 2021) but not a left boundary tone.

(2006) and Zhang (2014) propose that H% is a “segmentless” SFP and competes for the same syntactic position with segmental SFPs like *aa4* (at ForceP, or Tang 2015b’s “DegreeP”). Given that segmental SFPs occur before the RD chunk, we might expect H% to occur there as well.

- (61) (Ming4zai2 wui3 heoi3 Mei5gwok3 **aa4**)_l (*H%) ? (*aa4* cannot be pronounced as [a²⁵])
 Ming will go US SFP
 ‘Ming will go to the US?’

As first observed by Yip (2020), however, H% is severely degraded in gapped RD before the RD chunk. We extend this observation to gapless RD, including variants with imperfect copying, as illustrated in (62). In testing these cases, a pause should *not* be added before the RD chunk, which is also generally absent in RD (Liang 2002:76; Cheung 2009:198); though even with a pause added the placement of H% is still highly unnatural.¹⁶

(62) The ban on placement of H% in Cantonese RD

Context: We are discussing where Ming will go for studying abroad. You said he is applying for a US visa, and I ask:

- a. [wui3 heoi3 Mei5**gwok3** { *H% / **aa4** }] keoi5? (Gapped RD)
 will go US SFP 3SG
 ‘He will go to the US?’
- b. [Ming4zai2_j wui3 heoi3 Mei5**gwok3** { *H% / **aa4** }] Ming4zai2? (Gapless RD)
 Ming will go US SFP Ming
 ‘Ming will go to the US?’
- c. [Ming4zai2_j wui3 heoi3 Mei5**gwok3** { *H% / **aa4** }] keoi5? (Imperfect copying)
 Ming will go US SFP 3SG
 ‘Ming will go to the US?’

If RD constitutes one ι and there is no ι boundary before the RD chunk, the ban on H% in RD is expected. This effectively rules out the side-by-side two- ι phrasing (59b) in RD. One might, nevertheless, still entertain the possibility of recursive ι phrasing (59c) since H% at ForceP cannot be embedded and only occurs on the right edge of maximal ι .

We rule out the recursive ι phrasing by testing the Rising-Falling boundary tone HL%. As argued by Ki (2019), it occupies a lower syntactic position than *aa4* and H%.¹⁷ Interestingly, we observe that HL% may occur at the end of adverbial/subordinate clauses that form non-maximal ι , as in (63). HL% superimposes on the mid-level tone *gwok3* which is now pronounced as [k^wɔk³⁵³].

16. Boundary tones also cannot occur at the end of RD chunks (e.g., on the final *keoi5* in (62)), which is not surprising if they are segmentless SFPs that have the same syntactic distribution as segmental SFPs (i.e., banned at the end of RD, Cheung 2009). Yip (2020) also observes that it is possible to realize/“copy” H% twice on the end of both chunks, where a pause is needed between the two chunks. Yet, he shows that these occurrences should be characterized as two independent sentences (like afterthoughts, see §2) rather than RD due to their distinct syntactic and semantic properties, such as the presence of focus in the second/“RD” chunk, which would be infelicitous in the context in (62).

17. Based on co-occurrence constraints with other SFPs, she proposes that HL% occupies FocusP: [CoAP ... [Degree/ForceP H%/aa4 [FocusP HL% [CP ...]]]] (see Sybesma and Li 2007; Tang 2015b, 2020 for the full cartographic syntax of SFPs in Cantonese).

- (63) The rising-falling tone HL% may occur at right edges of non-maximal ι (*gwok* as [k^wɔk³⁵³])
- | | | | | | | | | | | |
|-------------------|-------------------------------|-----------|-------------|-----------------|-------|----------------------------------|-------------|--------------------------------|-----------|-------|
| [_{Root} | [_{CP_{adv}} | Seoi1jin4 | keoi5 | wui3 | heoi3 | Mei5 gwok3 ^{HL%} |], | [_{CP_{main}} | daan6hai6 | keoi5 |
| ι_{max} (| ι_1 (| | | | | |) ι_1 | ι_2 (| | |
| | | although | 3sG | will | go | US | | | but | 3sG |
| mei6 | jau5 | cim1zing1 |] |] | | | | | | |
| | | |) ι_2 |) ι_{max} | | | | | | |
- not.yet have visa
 ‘Although s/he will go to the US, but s/he hasn’t had a visa yet.’

What is crucial for us is that HL% cannot occur in RD before the RD chunks, as exemplified in (64). There are thus no right edges of non-maximal ι before the RD chunks, otherwise HL% should have been possible just like in (63).

- (64) The ban on placement of HL% in Cantonese RD
Context: Will s/he go to the US?
- a. Keoi5 wui3 heoi3 Mei5**gwok3**^{HL%}. (baseline, *gwok3* as [k^wɔk³⁵³])
 3sg will go US
 ‘(Of course) s/he will go to the US.’
- b. *[{___j/ Keoi5_j} wui3 heoi3 Mei5**gwok3**^{HL%}] keoi5_j. (Gapped/gapless RD)
 3sg will go US 3sg
 Int.: ‘(Of course) s/he will go to the US.’

To sum up this subsection, the failure to apply ι level phonology (boundary tone placement) in Cantonese RD supports the one-to-one ι phrasing of RD. In other words, the main chunk does not form a separate ι excluding the RD chunk, but an ι together with it.¹⁸

- $$(65) \quad \left[{}_{\text{CP1-main}} \left(\text{XP} \text{ YP SFP} \right) \right]]_t \quad \left[{}_{\text{CP2-RD}} \left[{}_{\text{DeFocP}} \text{ XP } \cancel{\text{YP SFP}} \right] \right]]_t$$
- ↑ **x** H%/HL%

Note that, however, the evidence here is only right-edge based. One could imagine a somewhat counterintuitive version of recursive ι phrasing where the RD chunk, despite being phonologically “lighter” (e.g., resisting stress, §2), serves as the head as in (66). Below, we will demonstrate that this possibility is ruled out in Mandarin, which has a juncture-sensitive tone sandhi. Although Cantonese does not have comparable left-edge-/juncture-sensitive processes, we will see from the acoustic evidence in §5 that this phrasing of RD does not exist in Cantonese too.

- (66) $\left[\begin{array}{c} \text{CP1-main (XP YP SFP)} \\ \left(\begin{array}{c} \text{CP2-RD [DeFocP XP ~~YP SFP~~]} \\ \left(\begin{array}{c} \text{ } \end{array} \right) \iota \end{array} \right) \iota_{max} \end{array} \right] \quad \begin{array}{l} \text{(to be rejected)} \\ \text{right-headed recursive } \iota \end{array}$

4.2 Third tone sandhi in Mandarin

Unlike Cantonese, boundary tone placement cannot be easily tested in Mandarin, since Mandarin question intonation is realized globally (as raising f_0 of the whole sentence, O. J. Lee 2005; Liu and Xu

18. While we do not provide examples of other boundary tones like H:% and L% due to space reasons, they also cannot occur before RD chunks.

2005), rather than locally at edges.¹⁹ However, Mandarin offers another type of phonological evidence for the lack of intonational phrasing in RD: third tone sandhi. T3 sandhi applies to consecutive T3 syllables, where the first one changes from a low tone to a rising tone, similar to the contour of tone 2 (Shih 1986, 1997; Duanmu 2000; M. Y. Chen 2000, *i.a.*), as in (67).

(67) Third tone (T3) sandhi in Mandarin

- | | | | |
|----|---|----|--|
| a. | T3-T3 → sT2 -T3
[²¹]-[²¹] → [³⁵]-[²¹]
L-L → LH -L | b. | jiu3-gui3 → jiu2 -gui3
[²¹]-[²¹] → [³⁵]-[²¹]
'alcoholic, lit. wine-ghost' |
|----|---|----|--|

As Shih (1997) observes, among many others, Mandarin T3 sandhi may apply across phonological phrase (ϕ) boundaries, such as a subject-VP juncture in (68). Yet, it does not apply across ι boundaries, such as a juncture between an adverbial clause and the main clause as in (69). The two ι are non-maximal in (69), showing that the ban on T3 sandhi is general to all ι junctures.

(68) T3 sandhi can apply across a subject-VP juncture

- [_{CP} Zuo2tian1 [_{Subj} na4-xiang1 shao1 **jiu3**^[21→35]] [_{VP} **shao3**^[21]-le0 yi1-ping2]]
 ϕ_1 (yesterday that-box Soju) ϕ_1 ϕ_2 (miss-PFV one-bottle) ϕ_2
 'Yesterday, one bottle of Soju went missing from that box of Soju.'

(69) T3 sandhi is not possible across clausal boundaries in complex sentences

- [_{Root} [_{CP_{adv}} Lao3wang2 shuo1 yao4 jin1tian1 **zou3**^[21/*35]], [_{CP_{main}} **ke3**^[21]shi4 mei2 zou3cheng2]]
 ι_{max} (ι_1 (Old Wang say want today leave) ι_1 ι_2 (but not left) ι_2) ι_{max}
 'Old Wang said that he wanted to leave today, but it didn't work out.' (Shih 1997:100)

When there is only one ι boundary, the application of T3 sandhi is also severely degraded, if not entirely blocked. For instance, T3 sandhi is highly unnatural across a left CP boundary of a complement clause of a verb in (70) and of a noun in (71).²⁰ By contrast, T3 sandhi is allowed across DP boundaries that correspond to ϕ boundaries, as exemplified in (72).²¹

(70) T3 sandhi is degraded across a left CP boundary (V complement clauses)

- [_{CP_{root}} Zhang1san1 cai1 **xiang3**^[21/??35] [_{CP_{comp}} **Lao3**^[21]wang2 ming2tian1 hui4 qu4 you2yong3]]
 ι_{max} (Zhangsan guess) ι (Old Wang tomorrow will go swimming) ι_{max}
 'Zhangsan guesses that Old Wang will go swimming tomorrow.'

19. We then expect that question intonation can occur in Mandarin RD, which is attested (Yip 2020).

20. A majority of our consultants found both sentences unnatural, but a few reported that (70) (n=2) and (71) (n=1) are acceptable if the sandhi tone is lengthened (i.e., "shallow-rising" in Lai and Li 2022). Importantly, they still found a contrast between these cases with ϕ -boundary cases (e.g., (68) and (72)) and RD cases (=74). For speakers who accepted T3 sandhi across subject clauses below in (68) (n=1), the same contrast is also found. See Kuang and Wang (2006) and Lai and Li (2022) for how varying phonetic realizations of the sandhi tone signal the strength of prosodic boundaries.

21. A disyllabic verb is chosen to avoid parsing monosyllables to the adjacent clauses. See Shih (1997) for the discussion that Mandarin foot formation involves "joining any remaining monosyllables to neighboring feet", leading to T3 sandhi between non-syntactic sisters like *gou3*^[21→35] *yao3* 'dog bites' (Subj Verb).

- (71) Tone 3 sandhi is degraded across a left CP boundary (N complement clauses)

$$[_{CP_{root}} Ta1\ mei2you3\ shuo1-q\dot{i}3^{[21/??35]}]_{\iota_{max}} [_{DP} [_{CP_{comp}} Lao3^{[21]}wang2\ xiang3\ li2kai1]\ de0\ shi4]]_{\iota_{max}}$$
 3SG not say-about Old Wang want leave DE matter
 ‘S/he didn’t mention the news that Old Wang wants to leave.’

- (72) Tone 3 sandhi can apply across a left DP boundary

$$[_{CP_{root}} Ta3\ mei3you3\ shuo1-q\dot{i}3^{[21/35]}]_{\iota} [_{DP} [Lao3^{[21]}wang2]\ de0\ shi4]]_{\phi}$$
 3SG not say-about Old Wang DE matter
 ‘S/he didn’t mention Old Wang’s issues.’

Similarly, T3 sandhi is degraded across a right CP boundary in (73) with subject clauses. This contrasts with (68) above, where T3 sandhi can apply across a DP subject.

- (73) Tone 3 sandhi is degraded across a right CP boundary (subject clauses)

$$[_{CP_{root}} [_{CP_{subj}} Lao3wang3\ ming2tian1\ hui4\ qu4\ you2yong3^{[21/??35]}]\ hao3^{[21]}xiang4\ bu4-ke3neng2]]_{\iota_{max}}$$
 Old Wang tomorrow will go swim seem im-possible
 ‘That Old Wang will go swimming tomorrow doesn’t seem possible.’

Now, we can test the juncture strength between the two chunks in RD by applying T3 sandhi. If there exists an ι boundary, we expect T3 sandhi to be at least degraded. As shown in (74), T3 sandhi is fully realized in RD, for both gapped and gapless variants. Imperfect copying patterns the same. This shows that the boundary in-between the two chunks is not as large as ι boundaries, but should be smaller, say, ϕ boundaries.

- (74) Tone 3 sandhi in Mandarin RD
Context: What does Old Wang like to do usually?
- a. [_ Xi3huan1 he1 **jiu3**^[21→35] Lao3^[21]wang2. (gapped RD)
 like drink wine Old Wang
 Lit.: ‘like drinking wine, Old Wang.’
- b. [Lao3wang2 xi3huan1 he1 **jiu3**^[21→35] Lao3^[21]wang2. (gapless RD)
 Old Wang like drink wine Old Wang
 Lit.: ‘Old Wang likes drinking wine, Old Wang.’
- c. [Ta1 xi3huan1 he1 **jiu3**^[21→35] Lao3^[21]wang2. (imperfect copying)
 3SG like drink wine Old Wang
 Lit.: ‘He likes drinking wine, Old Wang.’

Summing up, Mandarin provides phonological evidence from T3 sandhi for the prosodic phrasing of RD. The fact that T3 sandhi—a phonological process that applies at the ϕ -level or below—applies across the juncture between the main chunk and the RD chunk indicates the absence of both left and right ι -boundaries preceding the RD chunk. Rather than forming its own ι , the RD chunk is prosodically integrated into a single ι with the main clause, constituting a syntax-prosody mismatch.

- (75)
$$[_{CP1-main} (XP)\ YP\ SFP]]_{\iota} [_{CP2-RD} [_{DeFocP} XP\ \cancel{YP\ SFP}]]_{\iota}$$

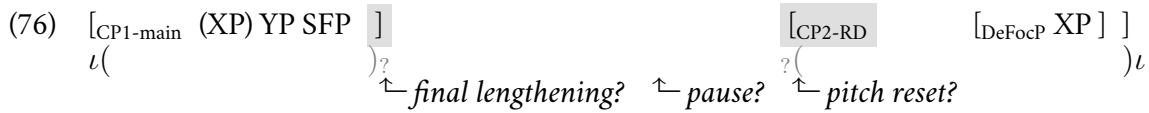
 ($\uparrow$ $\checkmark$ T3 sandhi $\rightarrow$)

5 Acoustic experiments on RD

In this section, we present two acoustic experiments that examine the prosodic cues available in RD, one for Cantonese and the other for Mandarin. Prosodic structures and their boundaries often have phonetic reflex cross-linguistically, including temporal cues like pre-boundary/final lengthening, tonal cues like pitch reset, and pauses (Edwards, Beckman, and Fletcher 1991; Wightman et al. 1992; Kawahara 2008, *i.a.*). We then expect the prosodic phrasing confirmed by *phonological* processes should also be reflected by corresponding *phonetic* cues. Moreover, some portion of the prosodic structure of RD is still underdetermined, including whether there is a non-maximal, left ι in Cantonese RD, and more generally, the internal sub- ι structure of RD in both languages. Since Cantonese and Mandarin do not have a rich set of phonological processes to diagnose different kinds of boundaries, prosodic cues shared by the two languages are useful in this regard.

To the best of our knowledge, no one has examined the prosody of RD in Chinese acoustically except Yip (2020) (for Cantonese and Mandarin) and Zhang (2022) (for Cantonese). While they conclude that RD consists of only one ι , consistent with our findings from phonological processes, they only examined gapped RD. Whether gapless RD behaves the same remains to be answered. This motivates the current study.²²

The experiments examine three prosodic cues that have been argued to signal ι boundaries in Cantonese (Chow 2005a, 2006, 2008; G. Li 2017) and Mandarin (Yang and Wang 2002; Chow 2005b, *i.a.*), as in (76). The three cues are: (i) final lengthening at the end of main chunks, which diagnoses right ι boundaries; (ii) pitch reset at the beginning of RD chunks, which diagnoses left ι boundaries; and (iii) pauses between main chunks and RD chunks, which signal ι juncture.



The two experiments share similar designs, jointly discussed in §5.1. The results are reported separately in §5.2 for Cantonese and in §5.3 for Mandarin, followed by an overall discussion in §5.4.

5.1 Design

Given the phonological evidence in the previous section, the experiments test the hypothesis below:

(77) Hypothesis of the experiments

RD patterns with canonical monoclausal sentences in having one intonational phrase, but not with canonical biclausal sentences, which have two intonational phrases. Specifically:

- a. The final syllable of the main chunk in RD does not have **final lengthening** comparable to the final syllable of the first clause in canonical biclausal sentences.
- b. The first syllable of the RD chunk does not have **pitch reset** comparable to the first syllable of the second clause in canonical biclausal sentences.

22. Moreover, Yip (2020)'s study did not examine final lengthening, and had a small sample size (n=6 for Cantonese; n=4 for Mandarin) without statistical tests. Zhang (2022) examined final lengthening and pitch reset with more subjects (n=20) and paired t-tests for data analysis. Nevertheless, her study had limited items of RD stimuli (n=3), and was not designed to contrast RD with genuine 2- ι sentences (but only with canonical, 1- ι sentences), which we have included in our design.

- c. The juncture between main and RD chunks does not have **pauses** comparable to the pauses at the juncture between the first and second clauses in canonical biclausal sentences.

To examine this hypothesis, the stimuli are constructed with a 2x2 factorial design, varied by *WordOrder* (Canonical vs. Dislocated) and *Length* (Short with 9 target syllables vs. Long with 11), as in Table 4. Canonical:Short (ShortCan) sentences are monoclausal, and Canonical:Long (LongCan) sentences are biclausal (i.e., juxtaposition of two ShortCan sentences). Dislocated:Short (ShortRD) sentences are gapped RD, and Dislocated:Long (LongRD) sentences are gapless RD. This gives in total 4 conditions, each condition with 12 lexical items. Each target sentence consists of syllables in the same tone (except for the SFPs in Mandarin which are toneless), and the 12 lexical items for each condition are evenly distributed over 4 tones in Cantonese (T1-T4, excluding T5-6)²³ and Mandarin (all the 4 tones, T1-T4); as well as declarative and interrogative clauses (i.e., each with 6 items). One example is illustrated for each condition in (78) in Cantonese (see Supplementary Material 1 for all the stimuli). To avoid utterance-initial/-final effects, each target sentence is preceded and followed by other phrases where the closest syllables to the target sentence always carry the same tone as the target syllables. Since Short conditions have 9 target syllables but Long conditions have 11, the preceding phrases of Short conditions are two-syllable longer than those of Long conditions.

	Short	Long
Canonical	[1] S _{σσ} Adv _{σσ} V _{σσ} O _{σσ} SFP _σ	[2] S _{σσ} Adv _{σσ} V _{σσ} O _{σσ} SFP _σ , S _{σσ} Adv V O SFP
Dislocated	[3] Adv _{σσ} V _{σσ} O _{σσ} SFP _σ S _{σσ}	[4] S _{σσ} Adv _{σσ} V _{σσ} O _{σσ} SFP _σ S _{σσ}

Table 4: A 2x2 factorial design (subscripted σ = target syllables)

(78) Sample stimuli by condition, illustrated in Cantonese (target syllables **bolded**)

[1].Canonical, Short (ShortCan)

(Lou5si1 m4-hai6 aa3.) **Sung3zi3 fong3gaa3 heoi3-gwo3 Taai3gwok3 ge3.** (Fu3zeon3 ...)
 teacher not-be SFP Sungzi on.holiday go-EXP Thailand SFP Fuzeon
 ‘(No, teacher.) Sungzi has been to Thailand on holiday. (Fuzeon ...)’

[2].Canonical, Long (LongCan)

(Ngo5 zi1 aa3.) **Sau3jin3 fong3gaa3 heoi3-gwo3 Taai3gwok3 ge3. Sung3zi3**
 1sg know SFP Saujin on.holiday go-EXP Thailand SFP Sungzi
 fong3gaa3 dou1 heoi3-gwo3 Taai3gwok3 ge3.
 on.holiday also go-EXP Thailand SFP
 ‘(I know.) Saujin has been to Thailand on holiday. Sungzi also has been to Thailand on holiday.’

23. Cantonese T5 low rising tone [13] and T6 low level tone [22] were not chosen because of two reasons. First, there are recent emerging tone mergers between (i) T2[35]-T5[13], (ii) T3[33]-T6[22], and (iii) T4[21]-T6[22], which are usually found in the younger generation (Bauer, Cheung, and Cheung 2003; Mok, Zuo, and Wong 2013, *i.a.*). To minimize individual variations due to tone mergers, T5 and T6 were excluded. Second, T5 and T6 SFPs are in general rare, particularly for outer SFPs (i.e., SFPs that are structurally higher) (Tang 2015b). Given that outer SFPs are usually required in Cantonese RD (Yip 2020), T5 and T6 were excluded to maximize the naturalness of experimental sentences.

[3].Dislocated, Short (ShortRD)

(Lou5sil m4-hai6 aa3.) **Fong3gaa3 heoi3-gwo3 Taai3gwok3 ge3 Sung3zi3.** (Fu3zeon3 ...)
 teacher not-be SFP on.holiday go-EXP Thailand SFP Sungzi Fuzeon
 ‘(No, teacher.) Has been to Thailand on holiday, Sungzi. (Fuzeon ...)’

[4].Dislocated, Long (LongRD)

(Ngo5 zi1 aa3.) **Sung3zi3 fong3gaa3 heoi3-gwo3 Taai3gwok3 ge3 Sung3zi3.**
 1SG know SFP Sungzi on.holiday go-EXP Thailand SFP Sungzi
 (Fu3zeon3 ...)
 Fuzeon
 ‘(I know.) Sungzi has been to Thailand on holiday, Sungzi. (Fuzeon ...)’

The experiment took place in a soundproof booth at Department of Linguistics, Yale University. In total, 26 subjects were recruited from New Haven, Connecticut, including 13 native speakers of Hong Kong Cantonese (age: 22-49, female: 7, male: 6) and 13 native speakers of Northern Mandarin (age: 18-29, female: 9, male: 4).²⁴ The subjects all self-reported that they had no history of speech, language, or hearing disorders. Each subject received a compensation of USD15.

Each subject participated in a session that lasted approximately 60 minutes, including instructions and breaks. There was a warm-up trial at the beginning with 6 (non-stimuli) sentences. A total of 72 sentences (48 target items + 24 fillers) were presented to the subjects in 4 blocks, each block containing 18 sentences. In each trial, one sentence was visually shown once at a time on a screen under an appropriate context (see Supplementary Material 2 for samples), and subjects were required to read aloud the sentence. The set of 18 sentences in a block was repeated 3 times in a randomized fashion. The subjects did not see the same lexical items in different conditions within a given block. There was a short break after each block. The ordering of the 4 blocks was (near-)balanced over the subjects. In total, 4 conditions x 12 lexical items x 3 repetitions x 13 subjects x 2 languages = 3744 tokens were obtained and analyzed. The sounds were recorded by a high-quality Shure microphone via Audacity (mono channel, sampling rate = 44.1kHz, resampled to 16kHz for forced alignment).

Three major prosodic properties of intonational phrases were measured, as illustrated in Figure 1 (for a ShortRD sentence) and (79). The first one, final lengthening, was measured by the duration (ms) of the SFPs (**bolded**). They are ι -final in ShortCan and LongCan. If the internal right boundaries in RD are ι -boundaries, we expect that the duration of SFPs in RD does *not* differ from ShortCan/LongCan, and otherwise if the boundaries are not ι -boundaries. The second one, pitch reset, was measured by the pitch (f0) of the syllables at right edges as compared to the pre-edge syllables. It is calculated by the subtraction of the maximal f0 of the 8th (Short) or 10th (Long) target syllable (underlined) from the maximal f0 of the 7th (Short) or 9th (Long) target syllable (underlined) in Hz ($\Delta f0 = \sigma_{8/10}$'s maxf0 - $\sigma_{7/9}$'s maxf0). The 8th syllable in ShortCan does not follow any prosodic boundary larger than a syllable (i.e., it forms a prosodic word, ω , with the preceding syllable). There should be no pitch reset, i.e., $\Delta f0$ should be zero or negative. On the other hand, the 10th syllable LongCan follows an ι boundary, and $\Delta f0$ should be a large value. If there is a left ι boundary before the RD chunk, we expect the $\Delta f0$ value of RD to be similar to LongCan's, but not ShortCan's; and otherwise if there is no such right ι boundary. The third one is pause, the duration (ms) of silence in between two edge-syllables, i.e., between the 7th and 8th target syllables in Short and between the 9th and 10th target syllables in Long (notated as #). To facilitate the measurement, the 8th syllable

24. Initially 14 Cantonese speakers were recruited, but one was not from Hong Kong and the data were discarded.

in ShortRD and the 10th syllable in Long all have a fricative onset, such that any silence before the frication can be easily observed. ShortCan is not controlled for as we do not expect any pause within ω .

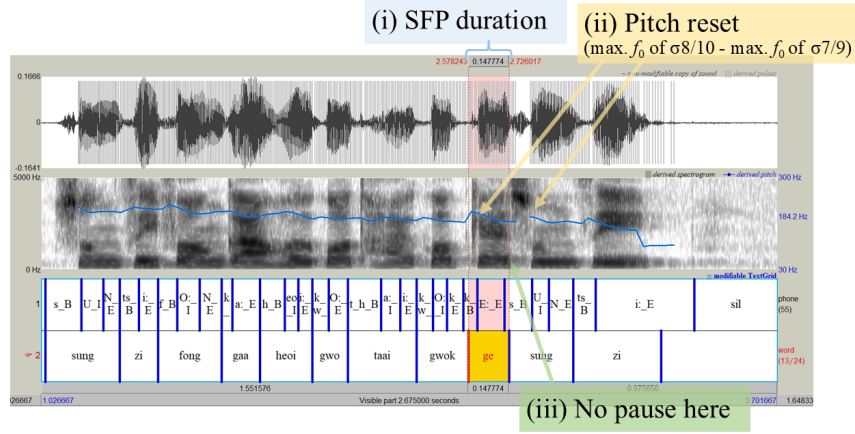


Figure 1: The three prosodic cues measured in a ShortRD sentence in Cantonese

(79) [1].ShortCan

ι -final SFPs; no boundary within ω

$\sigma_1 \sigma_2 \sigma_3 \sigma_4 \sigma_5 \sigma_6 (\sigma_7 \# \sigma_8)_\omega (\sigma_9)_\iota$

Subj Adv V-cplx Obj **SFP**

[2].LongCan

ι -final SFPs; ι boundaries

$\sigma_1 \sigma_2 \sigma_3 \sigma_4 \sigma_5 \sigma_6 \sigma_7 \sigma_8 (\sigma_9)_\iota \# \iota(\sigma_{10} \sigma_{11})$

Subj-1 Adv V-cplx Obj **SFP** [CP2 Subj-2

[3].ShortRD

X -final SFPs; unknown X boundaries

$\sigma_1 \sigma_2 \sigma_3 \sigma_4 \sigma_5 \sigma_6 (\sigma_7)_X \# X(\sigma_8 \sigma_9)$

Adv V-cplx Obj **SFP** [CP2 SubjDeFoc

[4].LongRD

Y -final SFPs; unknown Y boundaries

$\sigma_1 \sigma_2 \sigma_3 \sigma_4 \sigma_5 \sigma_6 \sigma_7 \sigma_8 (\sigma_9)_Y \# Y(\sigma_{10} \sigma_{11})$

Subj Adv V-cplx Obj **SFP** [CP2 SubjDeFoc

Acoustic measurements were carried out by using Praat (Boersma and Weenink 2017), supplemented by a Praat script ProsodyPro (Xu 2013). Syllables were forced-aligned by the Cantonese Forced Aligner (Lee and Tao 2021) for Cantonese, and by Charsiu: A transformer-based phonetic aligner (<https://github.com/lingjzhu/charsiu>) for Mandarin, with manual boundary adjustment for each item in Praat. In cases where a creaky voice intervenes in the f_0 analysis, pulses were added manually when feasible. f_0 smoothing was done by ProsodyPro.

As for the data analysis, syllable duration was $\log_{10}$ -transformed for obtaining normal distribution and then normalized to z-scores by participants (henceforth $z\logDuration$); and f_0 was also normalized to z-scores by participants (henceforth zf_0). We fitted linear mixed-effects models (LMMs, using the `lmer` function in the `lme4` package in R; Bates et al. 2015; R Core Team 2024) to the data in a stepwise fashion to examine the effects of *WordOrder* and *Length*, as well as their interaction. In all the models, *Participant*, *Lexical* (for the 12 items), and *Trial* (for the 3 repetitions) were included as random effects, but whether a by-*Participant* random slope was included was determined by model comparison, so as to examine potential individual differences.

5.2 Results for Cantonese

Final lengthening. LMMs were fitted to `zlogDuration` of SFPs. Fixed effect comparisons showed that adding *WordOrder* to a model with *Length* alone yielded a significant improvement in fit ($\chi^2(1) = 45.57, p < .001$), and inclusion of *Length* $\times$ *WordOrder* interaction yielded an additional improvement ($\chi^2(1) = 6.5, p = .001$). Random effect comparisons showed that adding by-*Participant* random slopes for *WordOrder* substantially improved fit relative to a random-intercept-only model ($\chi^2(2) = 83.8, p < .001$), which was not the case for inclusion of *Length* alone, *Length* and *Order*, or both with the interaction *Length* $\times$ *WordOrder* in the slopes ($p = .99; p = 1.00; p = .82$). The final model was: `zlogDuration ~ Length * WordOrder + (1 + WordOrder | Participant) + (1 | Lexical) + (1 | Trial)`.

The model revealed a robust main effect of *WordOrder*, such that the SFPs in RD were significantly shorter than those in Canonical ($\beta = -2.06, SE = 0.11, t(16) = -18.76, p < .001$). The main effect of *Length* was not significant ($\beta = -0.055, SE = 0.047, t(1827) = -1.182, p = .24$), but there was a significant yet small interaction between *Length* and *WordOrder* ($\beta = -0.169, SE = 0.066, t(1827) = -2.551, p = .011$), suggesting that reduction in duration under LongRD was slightly larger than under ShortRD. In other words, there was no final lengthening in RD comparable to ι -final syllables in Canonical. The differences in SFP duration are plotted in Figure 2 (in raw milliseconds for interpretability).

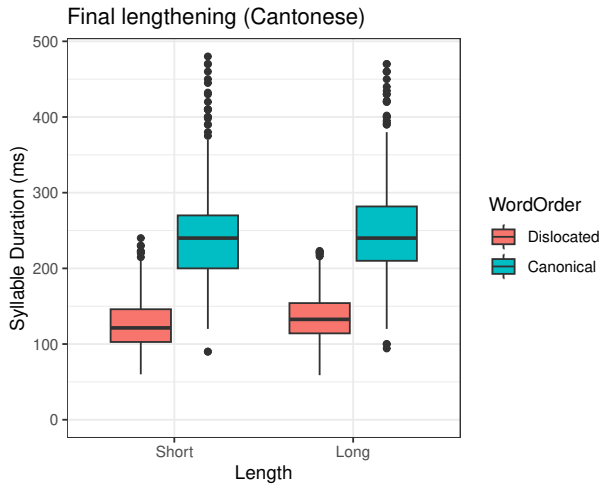


Figure 2: The duration of SFPs in Cantonese

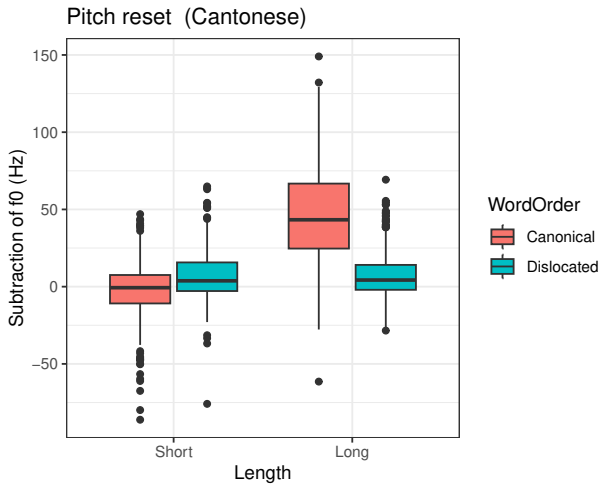


Figure 3: Pitch reset ($\Delta_{\max}f_0$) in Cantonese

Pitch Reset. LMMs were fitted to the subtraction of maximal normalized f_0 of the target syllables from the preceding ones ($\Delta z f_0$). Fixed effect comparisons showed that including *Length* $\times$ *WordOrder* interaction improved the model significantly as compared to models with only *Length* or no interaction ($\chi^2(1) = 47.59, p < .001$). Random effect comparisons showed that including random slopes for both *Length* and *WordOrder* as well as their interaction provided a significantly better fit than simpler models without the interaction in the slopes or with only random intercepts ($\chi^2(7) = 15.57, p = .029$). The final model was: `$\Delta z f_0 \sim \text{Length} * \text{WordOrder} + (1 + \text{Length} * \text{WordOrder} | \text{Participant}) + (1 | \text{Lexical}) + (1 | \text{Trial})$` .

There was a robust main effect of *Length*, such that Long sentences had larger $\Delta\text{zf}0$ values ($\beta = 1.71$, $SE = 0.08$, $t(13.1) = 22.78$, $p < .001$). The main effect of *WordOrder* did not reach significance ($\beta = -0.05$, $SE = 0.05$, $t(15.7) = -1.07$, $p = .30$). Crucially, however, a strong interaction between *Length* and *WordOrder* was observed ($\beta = -1.74$, $SE = 0.08$, $t(16.3) = -21.35$, $p < .001$). Post-hoc Tukey-adjusted pairwise comparisons confirmed that ShortRD and LongRD showed significantly lower $\Delta\text{zf}0$ than LongCan (both $p < .001$), but no such difference was found with ShortCan ($p = .73$; $p = .47$). We can conclude that RD did not have pitch reset that is comparable that of ι -initial syllables in LongCan. The differences are plotted in Figure 3 (again, in raw f0 for interpretability).

Although no differences between RD and *ShortCan* were found in the overall model, the model improvement of by-*Participant* random slopes for *Length* $\times$ *WordOrder* indicated individual differences. We therefore plot the raw $\Delta\text{f}0$ values for each subject in Figure 4. Interestingly, 8 out of 13 subjects (C01–C05 and C10–C12) showed a slightly larger value in RD than *ShortCan*, as confirmed by unpaired t-tests in `geom_signif`, indicating a small pitch reset in RD.

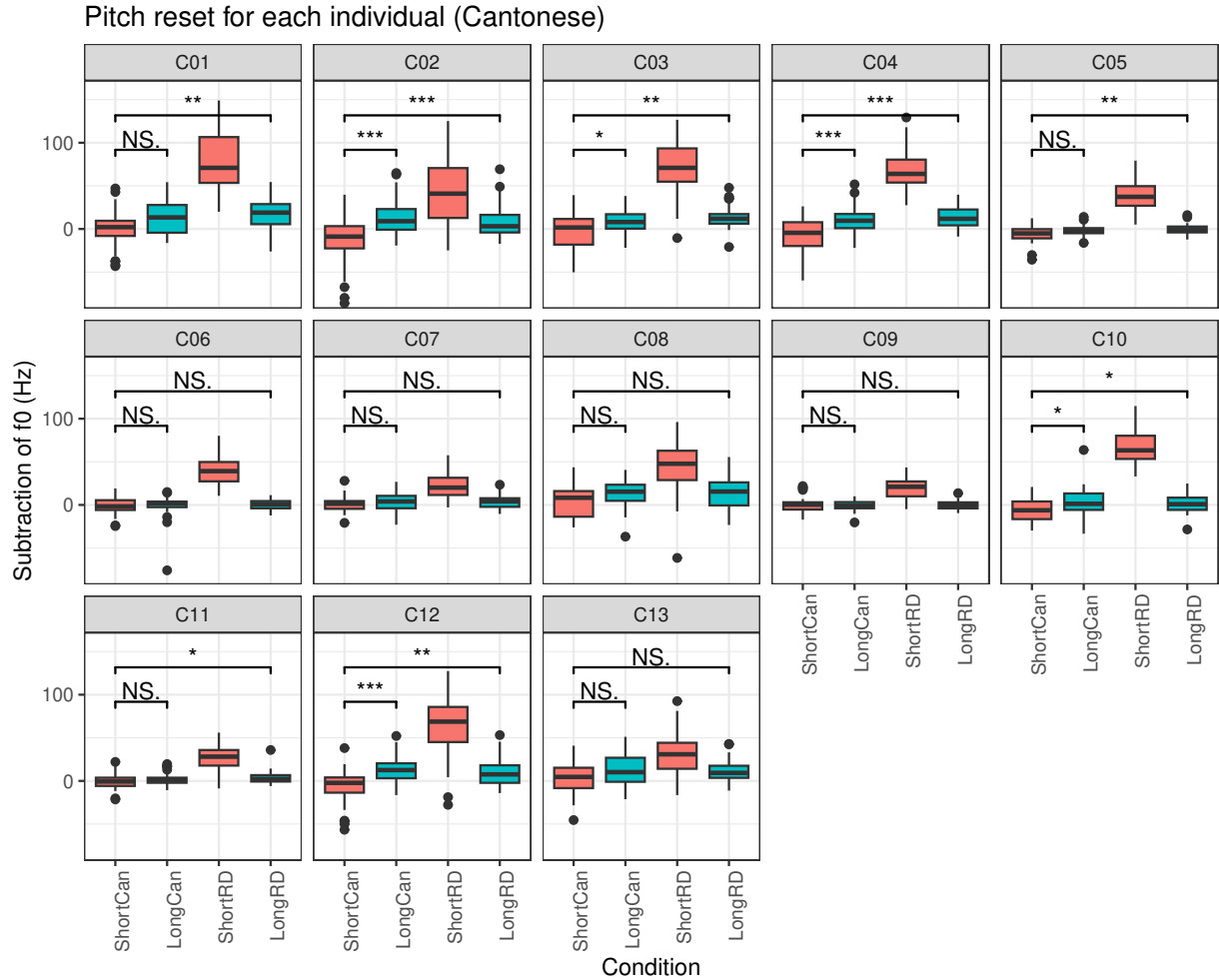


Figure 4: Individual differences in pitch reset ($\Delta\text{max.f}0$) in Cantonese

Pause. As expected, there were no pauses in ShortCan between its 7th and 8th syllables. All the tokens of LongCan contained a silence between the 9th syllable (end of CP1) and the 10th syllable (beginning of CP2), with a mean duration of 431 ms (range: 6–1480 ms, $SD = 212$). In contrast, only two tokens of LongRD (0.004%) exhibited a silence (42 ms and 96 ms). ShortRD also contained very few cases (6 tokens, 0.012%) with silence ($M = 64$ ms, range: 20–189 ms, $SD = 58$). Therefore, even without statistical tests, we can conclude that pauses were effectively absent in RD.

5.3 Results for Mandarin

Final lengthening. LMMs were fitted to $\logDuration$ of the SFPs. Including both fixed effects (*Length* and *WordOrder*) significantly improved model fit relative to a *Length*-only model ($\chi^2(1) = 51$, $p = < .001$), but not including their interaction did not ($p = 0.063$). The inclusion of by-*Participant* random slopes for *WordOrder* also significantly improved model fit compared to the random-intercepts model ($\chi^2(2) = 56.94$, $p < .001$), while adding slopes for *Length* alone, both *Length* and *WordOrder*, both with the *Length* $\times$ *WordOrder* interaction did not ($p = .62$; $p = .43$; $p = .70$). The final model was: $\logDuration \sim Length + WordOrder + (1 + WordOrder | Participant) + (1 | Lexical) + (1 | Trial)$.

There was a strong main effect of *WordOrder*, such that the SFPs in RD were significantly shorter than those in Canonical ($\beta = -1.64$, $SE = 0.064$, $t(13) = -25.38$, $p < .001$). The main effect of *Length* was not significant ($\beta = 0.026$, $SE = 0.024$, $t(1807) = 1.11$, $p = .27$). Like the case of Cantonese, there was no final lengthening in RD comparable to ι -final syllables in Canonical. The differences in SFP duration are plotted in Figure 5 (in raw milliseconds).

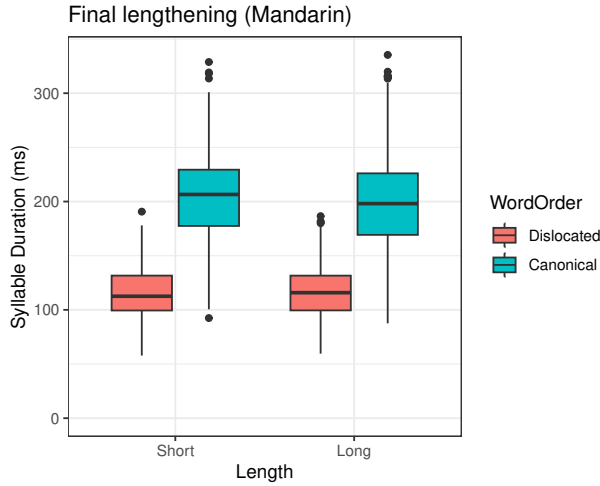


Figure 5: The duration of SFPs in Mandarin

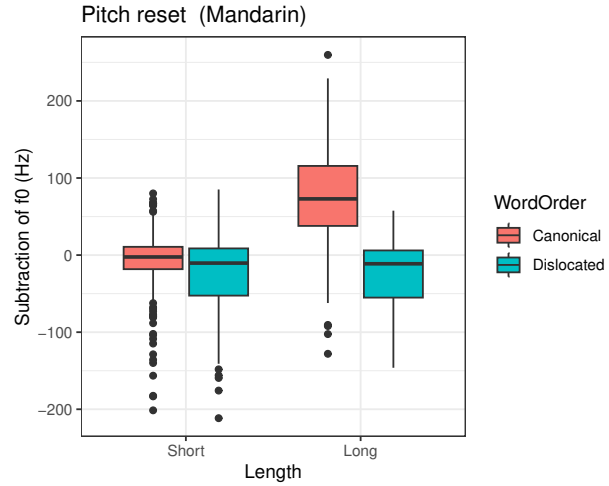


Figure 6: Pitch reset ($\Delta \max.f_0$) in Mandarin

Pitch Reset. LMMs were fitted to $\Delta z f_0$. Fixed-effects comparisons confirmed that adding *Length* $\times$ *WordOrder* significantly improved fit over simpler models with only *Length* or without the interaction ($\chi^2(1) = 36.74$, $p < .001$). Random effect comparisons showed that including random slopes for *Length* $\times$ *WordOrder* also significantly improved model fit relative to simpler random-effect structures ($\chi^2(7) = 44.8$, $p < .001$). The final model was: $\Delta z f_0 \sim Length * WordOrder + (1 + Length * WordOrder | Participant) + (1 | Lexical) + (1 | Trial)$.

$WordOrder \mid Participant) + (1 \mid Lexical) + (1 \mid Trial)$.

The model revealed significant main effects of both *Length* and *WordOrder*, as well as a strong interaction between them. The values of Δf_0 were higher in Long than Short sentences ($\beta = 1.54$, $SE = 0.10$, $t(13.0) = 15.46$, $p < .001$), and in Canonical sentences than in RD ($\beta = -0.31$, $SE = 0.08$, $t(13.2) = -4.06$, $p = .001$). There was a robust *Length* $\times$ *WordOrder* interaction ($\beta = -1.56$, $SE = 0.10$, $t(13.6) = -14.88$, $p < .001$), suggesting that the magnitude of *WordOrder* effects was much stronger in Long conditions than in Short conditions. Tukey-adjusted post hoc contrasts confirmed that both Canonical conditions exhibited greater pitch reset than their RD counterparts (all $p < .001$), while no difference was observed between the RD conditions ($p = .95$). LongCan showed significantly higher reset than ShortCan ones ($\beta = -1.54$, $SE = 0.10$, $t = -14.85$, $p < .001$). We can conclude that RD did not have a pitch reset comparable to the ι -initial syllable in LongCan, and, unlike Cantonese, even had a decrease in pitch as compared to the ω -internal syllables in ShortCan. The differences are plotted in Figure 6 (in raw f_0).

Again, individual differences were confirmed by the model improvement of by-*Participant* random slopes for *Length* $\times$ *WordOrder*. As shown in Figure 7 (in raw Δf_0 values for each subject), 7 out of 13 subjects (M01–M04, M08, M10, and M13) did not show a lower value for RD as compared to ShortCan, as examined by unpaired t-tests in `geom_signif`.

Pause. No pauses were observed in ShortCan or the RD conditions. Only LongCan had pauses, with a mean duration of 245 ms (range: 63–1141 ms; $SD = 187$).

5.4 Overall discussion

First, in Cantonese and Mandarin, the SFPs in both gapped and gapless RD have a significantly shorter duration than the SFPs in canonical sentences, which are on the right edge of ι -boundaries and undergo final-lengthening (e.g., Chow 2008 for Cantonese and Yang and Wang 2002 for Mandarin). This supports the findings by Zhang (2022) on Cantonese gapped RD. In other words, SFPs in RD do not have final lengthening of the sort on right ι -boundaries, suggesting that there are no such right ι -boundaries at the end of main chunks. This corroborates the conclusion from phonological processes in both languages in §4.

Second, no large pitch reset comparable to the one in LongCan is found in RD in both languages, the latter of which represents a left ι -boundary. This suggests the absence of such a boundary at the beginning of RD chunks. Effectively, this rules out the alternative right-headed recursive ι phrasing in Cantonese, and is also consistent with the fact that T3 sandhi in Mandarin can apply across RD chunks. Note that the two languages differ in whether RD chunks have lower Δf_0 values than ShortCan, where the target syllable is within a prosodic word and does not follow any prosodic boundary. In Cantonese, it is not the case, which can be understood as the total absence of pitch reset in RD; but in Mandarin it is the case, and the values are indeed usually negative (median below 0, as can be seen in Figure 6), suggesting a further lowering of the pitch on RD chunks. The difference was also noticed by Yip (2020). This could be due to different uses of f_0 cues in the two languages. It is known that Cantonese does not readily employ f_0 cues for focus, including f_0 boost, f_0 range expansion, and post-focal reduction (Wu and Xu 2010; Fung and Mok 2014; Hsu, Xu, and Ngai 2018; Yang 2022, *pace* Man 1999; Gu and Lee 2007), which Mandarin recruits for marking focal prominence and post-focal non-prominence (e.g., Chen, Wang, and Xu 2009; Xu, Xu, and Sun 2004; Xu, Chen, and Wang 2012). Since RD chunks are always defocused, their f_0 may be affected by focus prosody in Mandarin, in

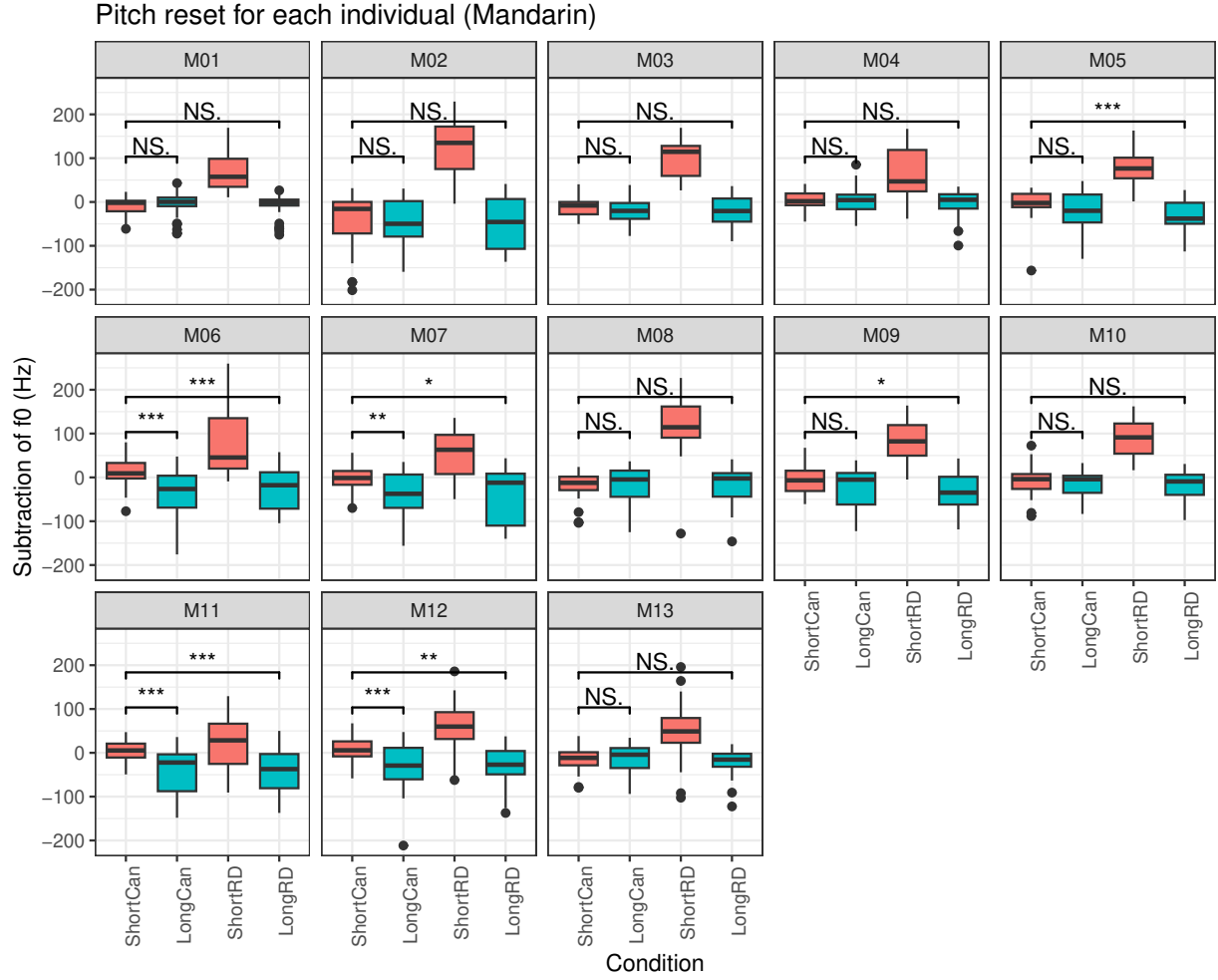


Figure 7: Individual differences in pitch reset ($\Delta_{\max}f_0$) in Mandarin

addition to prosodic boundaries. Put differently, the lowered pitch is plausibly a way to signal non-prominence in Mandarin.

Interesting individual differences in pitch reset were observed in both languages. In Cantonese, over 60% of the speakers have a slight yet significant pitch reset in RD (as compared to ShortCan); whereas in Mandarin, over half of the speakers do not have significant pitch lowering. Zhang (2022) also found a slight pitch reset for T4 gapped RD in Cantonese. If we take the degree of pitch reset to be an indication of the relative strength of the prosodic boundary (e.g., G. Li 2017 for Cantonese, Yang and Wang 2002 for Mandarin), we can explain the slight pitch reset by positing a phonological phrase (ϕ) boundary before RD chunks in Cantonese. ϕ -boundaries are weaker than ι , and thus the degree of pitch reset is lower. As for Mandarin, assuming that RD chunks are always f_0 -reduced due to independent factors (e.g., defocus/non-prominence), the lack of such f_0 reduction might in turn suggest some sort of ϕ -phrasing, whose effects on pitch reset are “canceled out” by focus prosody. As for the speakers who did not share the above patterns, a possible explanation is that they parsed the RD chunks not as a ϕ but as a ω , hence the absence of pitch reset (for Cantonese), and the f_0 may be

further lowered due to focus prosody (in Mandarin). We will entertain this possibility in §6.2.²⁵

Third, pauses are basically absent before RD chunks in both languages, which are always present in LongCan to signal ι junctures. This further reinforces that there are no ι junctures between the two chunks in RD. Together with final lengthening and pitch reset, we can conclude that none of the prosodic cues support the existence of internal ι boundaries in RD. Hence, RD consists of only one ι , and the boundaries before RD chunks are at the strongest a ϕ boundary (possibly weaker), as in (80). To recap, we see a case of syntax-prosody mismatch: two CPs, but one ι .

- (80)
$$\begin{array}{ccccccc} \text{[CP1-main (XP) YP SFP]} & & & & \text{[CP2-RD]} & & \text{[DeFocP XP]} \\ \iota(& & \phi & & \phi & &)\iota \\ & \nwarrow \text{final lengthening?} & \nwarrow \text{pause?} & \nwarrow \text{pitch reset?} & & & \\ & \text{NO} & \text{NO} & \text{slight/none} & & & \end{array}$$

6 Defocus Rephrasing and syntax-prosody mismatches

We have seen a case of syntax-prosody mismatches in RD in Cantonese and Mandarin, as schematized in (81) (reproduced from (58)). What is puzzling about this case is that both languages, while having drastically different acoustic properties in focus prosody, share two patterns: (i) focus does not trigger phonological rephrasing (Wu and Xu 2010; Zhang, Clayards, and Wagner 2024; also §7), in lines with the NO-REPHRASING view; but nevertheless, (ii) the mismatch seems to be related to focus in the sense that RD chunks are defocused, apparently at odds with the NO-REPHRASING view.

- (81) The prosodic mismatch in of RD (only ι shown): Two CPs, one ι
- | | | |
|--|---|---------------------------|
| $\begin{array}{c} \text{[CP1-main (XP) YP SFP]} \\ (\end{array}$ | $\begin{array}{c} \text{[CP2-RD [DeFocP XP YP SFP]} \\)\iota \end{array}$ | <i>syntax
prosody</i> |
|--|---|---------------------------|

We suggest that this case reveals the role of defocus in prosody. As we will propose in §6.1–6.2, the defocus nature of the RD chunks (indirectly) triggers the mismatch. The leading idea is that defocus, unlike focus, cannot receive *head* prominence. As a result, if the RD chunk (a clause/CP) were mapped onto an intonational phrase, the only overt material, namely the defocus, could not serve as the prominent prosodic head. To avoid forming an illegitimate headless intonational phrase, the RD chunk is integrated into the preceding intonational phrase formed by the main chunk, hence giving rise to the mismatch. The proposed analysis represents a DEFOCUS REPHRASING view, which distinguishes the role of defocus from focus in prosody. Consequently, even though focus does not trigger mismatches, defocus does, as in the case of Cantonese and Mandarin and to be discussed in §7.

6.1 An Optimality-Theoretic analysis

Couched within an Optimality-Theoretic approach to the syntax-phonology interface (Selkirk 1986; Truckenbrodt 1995, *i.a.*), we assume Selkirk (2011)’s Match Theory where the mapping of syntactic

25. Another possible explanation is to keep the prosodic phrasing with ϕ constant, but assume that individuals may vary in the “weighting” of different cues when realizing the same type of boundaries. For example, some may prefer using f0 cues, realized as an intermediate pitch reset; and some prefer other cues over f0 cues, hence the pitch reset is not observed. Potentially, this possibility can be tested with a more comprehensive design that takes all kinds of cues into consideration (e.g., articulatory strength, intensity, etc.), and see whether there are trade-off relations between different cues (cf. Kim 2020 for relevant discussions in English). We leave this possibility to future research.

structure to prosodic domains is governed by a set of violable constraints.²⁶ Other (morpho)phonological constraints may participate in the calculation of the optimal candidate and outrank the mapping constraints, creating syntax-prosody mismatches. In this way, mismatches are captured in a predictable manner. Three core constraints are involved: (i) a mapping constraint $\text{MATCH}(\text{CP}, \iota)$, (ii) a headedness constraint $\iota\text{:HEAD}$, and (iii) a novel constraint DEFOCUS , which we will formulate one by one below with clarifications on the theoretical premises.

First, we assume the $\text{MATCH}(\text{CP}, \iota)$ constraint in (82). When there is no conflict with other higher-ranked constraints, (82) ensures an isomorphic syntax-prosody mapping.

- (82) $\text{MATCH}(\text{CP}, \iota)$ (after Selkirk 2011:451)
Assign a violation mark if the left and right edges of a CP in the input syntactic representation do not correspond to the left and right edges of an intonational phrase in the output phonological representation.

The notion of clausehood has been subject to debate. One line in the literature takes CP, a syntactic phase (Chomsky 2001), to be the locus (Cheng and Downing 2009, 2012). Another line, however, takes speech act and its relevant projection (e.g. SAP/ForceP) as the locus (e.g., Selkirk 2005; Truckenbrodt 2015; see Selkirk 2009 for a hybrid view). Clausehood here thus refers to “illocutionary clauses”, which are by default root clauses (B. T. Downing 1970; Emonds 1970).²⁷ We follow the first line and suggest that, at least in Cantonese and Mandarin, it is CP that maps onto ι but not just the root clause/illocutionary clause. One piece of evidence comes from the blocking of T3 sandhi in Mandarin subordinate clauses, which are full CP clauses but not illocutionary clauses. As we have seen in §4.2, adverbial clauses, relative clauses, subject clauses, and complement clauses are mapped onto a prosodic domain large enough to block T3 sandhi, which is an ι instead of a ϕ (e.g., formed by DPs or VPs). Turning to Cantonese, boundary tones such as Rising-Falling tone HL% may also occur at the right edge of adverbial clauses, as we witnessed in §4.1. Therefore, non-illocutionary CPs in both languages are mapped to ι .

To give an illustration, the tableau (83) lists three output candidates of a biclausal input with two CPs, with subscripted i and j indicating correspondence. In candidate (a), the left and right edges of CP1 correspond to ι_i , so as CP2 to ι_j . Thus, $\text{MATCH}(\text{CP}, \iota)$ is not violated. In (b), however, the two CPs are parsed into one ι , and there is no ι matching CP1 or CP2, violating $\text{MATCH}(\text{CP}, \iota)$ twice. In (c), only CP1 has a matching ι . CP2 is unparsed and has no matching ι , violating $\text{MATCH}(\text{CP}, \iota)$ once.

- (83) A sample evaluation of $\text{MATCH}(\text{CP}, \iota)$

$[\text{CP}_1 \text{ XP YP}]_i [\text{CP}_2 \text{ ZP}]_j$	$\text{MATCH}(\text{CP}, \iota)$
a. $(\text{XP YP})_{\iota_i} (\text{ZP})_{\iota_j}$	
b. $(\text{XP YP ZP})_{\iota}$	*!*
c. $(\text{XP YP})_{\iota_i} \text{ZP}$	*!

The second constraint is $\iota\text{:HEAD}$ in (84), concerning the headedness of intonational phrases.

26. The proposed analysis below is also compatible with other versions of OT-style mapping constraints, such as ALIGN-XP (Selkirk 1986, 1995) or WRAP-XP (Truckenbrodt 1999), with extension to the clausal domain.

27. There are also proposals on flexible mapping, where the syntactic locus of ι varies with the height of verb movement (e.g. in Basaa (Bantu) and Hungarian, Hamlaoui and Szendrői 2015, 2017). In Chinese, verbs only move to ν/Asp below T (see Lee and Yip 2025 and references therein), and $\nu\text{P}/\text{AspP}$ arguably only forms ϕ instead of ι (e.g., no ι boundary after the SpecTP subject to block T3 sandhi in Mandarin). Note also that we follow Cheng and Downing (2012) and assume that it is the whole phase that is mapped onto a prosodic domain, not the complement of phase heads.

(84) IntonationalPhrase:Head (ι :HEAD)

An intonational phrase must have at least one daughter constituent designated as its head.

Prosodic heads here refers to an abstract constituent that is marked as strong/headed in a certain prosodic domain, encoding “an abstract notion of prominence” (Elordieta and Selkirk 2018:2). In the case of (non-recursive) ι , a ϕ , being one of the daughter constituents, is assigned as the head of the ι , such as the rightmost underlined ϕ_3 in candidate (a) in the tableau (85). A headless ι , such as candidate (b), violates ι :HEAD.

(85) A sample evaluation of ι :H

[_{CP} ZP YP XP]	ι :H
 a. ($\phi_1 \phi_2 \underline{\phi_3}$) $_{\iota}$	
b. ($\phi_1 \phi_2 \phi_3$) $_{\iota}$	*!

There are three notes on headedness in order. First, a prosodic head may or may not have phonetic realization like metrical prominence annotated as grid marks. In stress languages like English, head prominence has phonetic reflex as nuclear stress (associated with, e.g., a higher pitch accent and increased intensity) assigned within the rightmost ϕ_3 in canonical, focus-neutral sentences. In tonal languages like Mandarin, while usually assumed to lack English-type lexical stress accent, Feng (2019) argues that the effects of nuclear stress are still observable in various word order alternation phenomena related to prosodic weight (e.g., number of syllables), such as heavy NP shift, dative alternation, object *ba*-preposing, etc., where the phonologically heaviest constituent has a strong preference to be placed at the end. His arguments apply to Cantonese.²⁸ Moreover, Cantonese also has boundary tones that must be anchored to the right edge of ι , which we take to be a further indication of headedness on the ι -level. Note that we will need another constraint to align the head to the right edge, to be addressed in §6.2.

Second, ι :HEAD is crucially different from its antecedent, Selkirk (1996)’s HEADEDNESS in (86) as part of the Strict Layering hypothesis, in *not* specifying the prosodic category of the head.

(86) HEADEDNESS (as an undominated constraint) (Selkirk 1996, ex.4ii)
Any C^i must dominate a C^{i-1} (except if $C^i = \sigma$).

28. For example, Feng (2019) proposes that the contrast in example (i) in Mandarin (which dates back to Chao 1968) is due to the inability of a monosyllabic verb to receive nuclear stress at VP right edges after the object is preposed with *ba*, whereas a verb with more than one syllable can. Cantonese behaves the same as in (ii) with *zoeng* (= *ba*’s counterpart). We do not replicate all his arguments for space reasons.

(i) a. *Ni yinggai ba [yifu] **xi** $_{\sigma_1}$. [Mandarin]
2SG should BA clothes wash
‘You should wash your clothes’ (Feng 2019:193)

b. Ni dei xian ba [qingxing] **diao** $_{\sigma_1}$ **cha** $_{\sigma_2}$, zai ba [wentu] **fen** $_{\sigma_1}$ **xi** $_{\sigma_2}$, (...) (Feng 2019:195)
2SG should first BA condition investigate next BA problem analyze
‘You must first investigate the conditions, next analyze the problem, (...)’

(ii) a. Nei jinggoi zoeng [di-saam] { ***sei** $_{\sigma_1}$ / **sei** $_{\sigma_1}$ -**gon** $_{\sigma_2}$ **zeng** $_{\sigma_3}$ }. [Cantonese]
2SG should ZOENG CL.PL-clothes wash wash-clean
‘You should wash your clothes (clean).’

b. Nei jiu zoeng [cingfong] citdai { ***caa** $_{\sigma_1}$ / **diu** $_{\sigma_1}$ **caa** $_{\sigma_2}$ }, sin hoji wandou zansoeng.
2SG should ZOENG condition thoroughly investigate investigate, then can find truth
‘You must thoroughly investigate the conditions, then you can find the truth.’

(86) entails two requirements: (i) every prosodic category must be headed; and (ii) every prosodic category must dominate an (immediate) lower-level prosodic category (on the Prosodic Hierarchy) as the head. As an undominated constraint, these requirements are universal. Selkirk’s Headedness, however, is too strong and rules out recursive prosodic categories, in addition to headless categories. Given the robust evidence on prosodic recursion (Wagner 2005, 2010; see also Ito and Mester 2009; Weber 2022 for ω -recursion; Ito and Mester 2012, 2013 for ϕ -recursion; Ladd 1986, 2008 for ι -recursion), (84) retains the requirement that ι must be headed, but it imposes no restriction on the category that heads ι . Thus, recursive ι (i.e., ι_{max} headed by ι) is possible.²⁹

Third, ι :HEAD does not exclude the possibility of multi-headed categories, unlike other previously proposed headedness constraints. For example, Elordieta and Selkirk (2018)’s headedness constraint on prosodic word, ProsodicWord:Head [ω :Hd], specifies that heads must be unique. Similarly, Ishihara (2011), in accounting for Japanese focus prosody, dictates that “one and only one grid-mark [i.e., prominence]” can be assigned to ϕ and to ι in his CULMINATIVITY constraint. Conceptually, however, there is no *a priori* reason to assume the *existence* of heads must be associated with the *uniqueness* of heads, and there are also versions of headedness constraints that differentiate the two. For example, Hyman (2006) distinguishes OBLIGATORINESS (i.e., at least one head) from CULMINATIVITY (i.e., at most one head) in his proposal of stress accent on the ω -level. Hyman emphasizes that CULMINATIVITY must not be conflated with OBLIGATORINESS, since there are languages that do not observe CULMINATIVITY yet observing OBLIGATORINESS (e.g., Creek). Following his insight, the proposed ι :HEAD constraint in (84) only requires OBLIGATORINESS on the ι -level, and we leave whether CULMINATIVITY must apply to the ι -level as an open empirical issue.

We now turn to the last constraint, DEFOCUS in (87). We propose that defocus must not receive *head* prominence. Df refers to the element with the [–Foc] syntactic feature, which triggers movement of Df to SpecDeFocP (i.e., the RD chunk *excluding* the elided materials, see §2–3).

(87) DEFOCUS (Head prominence-based)

Let Df be a defocus element and DfP be the highest prosodic constituent in the output corresponding to Df. Assign a violation mark if DfP is a prosodic head and a daughter of a higher prosodic category or a higher projection of the same category as DfP.

The intuition is that while focus receives prominence and serves as a prosodic head (Truckenbrodt 1995), a *defocus* can never do so. We formulate the kind of prominence that defocus cannot receive as the abstract prominence encoded in a prosodic *head*. That is, defocus is always non-prominent since it is always a non-head. The prosodic reflex of non-prominence can be deaccenting in languages that assign heads a pitch accent. Such deaccenting effects are what we observed for Mandarin RD chunks from the acoustic results. In Cantonese, where both accenting and deaccenting are absent, the effects are manifested as a restriction on the number of syllables, such as Cheung (2015)’s heaviness constraint in gapless RD, stating that RD chunks are preferred to be shorter than main chunks.³⁰

Two illustrations are given below, one concerning the ι -level, another concerning the ϕ -level. In

29. Another difference is that Selkirk (1996) treats Headedness as an axiom-like constraint, i.e. it is undominated in constraint ranking, whereas ι :HEAD here is treated as a violable constraint. Nevertheless, ι :HEAD is presumably ranked very high in Cantonese and Mandarin, such that violations are not observed. We leave open whether the proposed ι :HEAD can be violated in other languages. For the existence of potential headless categories such as prosodic words, see Ito and Mester (2016) for unaccented loanwords in Japanese.

30. Zhang (2022) also found that the average duration of non-sentence-final syllables in Cantonese RD chunks is shorter than those in the main chunks (with marginal significance).

(88), the rightmost ϕ_k is the highest prosodic constituent in the output corresponding to Df, i.e. it is the DfP. In candidate (b), ϕ_k is assigned as the head of a higher prosodic category, namely ι , violating DEFOC. In contrast, in candidate (a), the head is another constituent, the middle ϕ . DEFOC is hence not violated, since the ϕ_k does not bear ι -headhood.³¹

(88) A sample evaluation of DEFOC on the ι -level

[_{CP} XP YP ZP _{Dfk}]	DEFOC
☞ a. ((XP) _{ϕ} (YP) _{ϕ} (ZP) _{ϕ_k}) _{ι}	
b. ((XP) _{ϕ} (YP) _{ϕ} (ZP) _{ϕ_k}) _{ι}	*!

(where ι 's prosodic head is underlined)

The same can be said to recursive ϕ in (89). In candidate (b), ϕ_k , the DfP, is the head of the recursive ϕ_{max} . Since ϕ_{max} is a higher projection of the same category as the DfP, DEFOC is violated. In order to avoid DEFOC violation, the left ϕ serves as the head, as in candidate (a).³²

(89) A sample evaluation of DEFOC on the ϕ -level

[YP ZP _{Dfk}]	DEFOC
☞ a. ((XP) _{ϕ} (ZP) _{ϕ_k}) _{ϕ_{max}}	
b. ((XP) _{ϕ} (ZP) _{ϕ_k}) _{ϕ_{max}}	*!

(where maximal ϕ is indicated by a subscripted $_{max}$, and ϕ_{max} 's prosodic head is **bolded**)

It is very important to note that DEFOC is different from constraints that result in deaccenting discourse-given phrases (e.g., Féry 2013's Destress-Given and Kratzer and Selkirk 2020's Dephrase-Given). Kratzer and Selkirk (2020) offer compelling arguments that the notion of contrastiveness and discourse givenness must be distinguished semantically, syntactically, and prosodically (see also Katz and Selkirk 2011 for evidence that contrastive focus has different acoustic correlates from discourse-new elements). The notion of focus adopted in this paper is based on Rooth (1992)'s alternative focus, which corresponds to their contrastiveness, including contrastive focus, *wh*-phrases, focus associated with operators like 'only' and 'even'; but not informational focus or givenness. As far as we can see, mere discourse givenness plays no role in Cantonese and Mandarin RD (see §2).

With the three core constraints in place, we can now account for the syntax-prosody mismatch in RD. We propose that the mapping constraint MATCH(CP, ι) is outranked by DEFOC and ι :H in Cantonese and Mandarin, as illustrated in (90).

(90) The proposed ranking of the three constraints in Cantonese and Mandarin

{DEFOC, ι :H} » MATCH(CP, ι)

With the proposed ranking, an output candidate of RD with syntax-prosody mismatches wins over those without, as illustrated in the tableau (91) below. If CP2 is mapped onto a ι_j without any daughter constituent assigned as the head, as in (a), ι :H is violated. If the head is assigned to the only overt material in CP2 (i.e., ZP), as in (b), the higher-ranked DEFOC is violated. It is because the DfP

31. This is only a sample evaluation. Middle-headed ι indeed violates right-headedness on ι in Chinese (see §6.2).

32. Note that, however, it is Df whose corresponding output cannot serve as a prosodic head of other projections, but there can be prosodic heads *inside* the corresponding output of Df. If, for example, Df contains two prosodic words ω , one ω can be the head (underlined):

(i) [_{ZP} A B]_{Df}
 (ω ω) _{ϕ}

ϕ_k that corresponds to the defocus, ZP, now heads a higher prosodic category ι_j . To avoid violating ι -headedness and defocus's non-head requirement, the syntax-prosody mapping “compromises”. In candidate (c), there is only one ι , and there is no ι corresponding to both CP1 and CP2 respectively, violating the lowered-ranked MATCH(CP, ι) twice. Yet, it does not violate ι :H and DEFOC—the ι has a head, which is the rightmost recursive ϕ_{max} ; moreover, the heads of both ι and ϕ_{max} are *not* assigned to the defocus (i.e., the DfP ϕ_k). Since it is less costly to violate the mapping constraint (i.e., less costly to have syntactic-prosody mismatches), candidate (c) is chosen. This captures the attested syntax-prosody mismatches in RD.³³

(91) Rephrasing triggered by headless ι

$[\text{CP}_1 \text{ ZP YP}]_i [\text{CP}_2 \text{ ZP}_{\text{DfP}} \text{ YP}]_j$	ι :H	DEFOC	MATCH(CP, ι)
a. $((\text{ZP})_\phi (\text{YP})_\phi)_{\iota i} ((\text{ZP})_{\phi k})_{\iota j}$	*!		
b. $((\text{ZP})_\phi (\text{YP})_\phi)_{\iota i} ((\text{ZP})_{\phi k})_{\iota j}$		*!	
c. $((\text{ZP})_\phi ((\text{YP})_\phi (\text{ZP})_{\phi k})_{\phi_{max}})_{\iota}$			**

(where ι 's prosodic head is underlined, and ϕ_{max} 's prosodic head is **bolded**)

Nevertheless, there are different ways to integrate the RD chunk (CP2) into the main chunk (CP1) prosodically. Candidate (c) above represents “deep integration”, where the RD chunk is parsed with the rightmost phonological phrase in the main chunk, forming a recursive ϕ . CP2 is fully integrated into CP1 and forms one ι as a whole. There are also other logically possible ways with different degrees of integration, as will be discussed in the next subsection.

Before proceeding, we would like to address a potential alternative source of rephrasing. The RD chunk also contains elided materials (via sluicing), in addition to the defocused elements. One may question whether it is the sluicing itself that leads to the rephrasing, presumably by deleting syntactic structures that should have been mapped onto ι (see, e.g., M. Chen 1987; Truckenbrodt 1999 for the view that silent materials are irrelevant to mapping). We argue against such a possibility. As shown convincingly in the series of work by D. Wu (2022a, and citations below), ellipsis in English does not alter prosodic boundaries. She demonstrates that the two answer strings in (92), while having the same segmental content, differ in prosody (D. Wu 2025a):

(92) Q: Where did Loretta go? (D. Wu 2025a:ex.5,7)

A1. [~~She went to~~ the restaurant] and [~~she went there~~ at midnight]. (Clausal conjuncts)

A2. ~~She went to~~ [the restaurant] and [the giftshop]. (DP conjuncts)

Specifically, the final rime (i.e., *restaurant*) and pause before *and* are significantly longer in A1 than in A2, signaling a larger prosodic domain mapped onto from the clausal conjunct. Similar experimental findings have been replicated for other elliptical constructions in English such as *either...or* sentences (D. Wu 2022a, 2022b) and corrective *but* sentences (D. Wu 2025b). Therefore, syntactic boundaries are still active in the mapping to prosody even after ellipsis.

Moreover, afterthoughts like (93), while arguably also involve sluicing internally, are mapped onto a separate ι (e.g., in Germanic languages, Ott and de Vries 2016).

33. For simplicity and readability, we do not represent the optional SFPs in (91) (see Lu 1980, Cheung 1997:7; Luke 2000:305; Y. Chen 2016:82; and Yip 2020 for SFP-less RD). SFPs are enclitics and attach to the closest preceding prosodic unit (i.e., ϕ here). The prosodic phrasing of an RD sentence with SFPs is represented below:

(i) $[\text{CP}_1 \text{ ZP YP SFP}]_i [\text{CP}_2 \text{ ZP}_{\text{DfP}} \text{ YP-SFP}]_j$
 $((\text{ZP})_\phi ((\text{YP})_{\phi \text{-SFP}} (\text{ZP})_{\phi k})_{\phi_{max}})_{\iota}$

- (93) a. I met a star today: **John Travolta!** (Ott and de Vries 2016:643)
 b. [CP₁ I met a star_i today] [CP₂ John Travolta_i ~~I met t_i today~~]!

There are various grammatical and discourse differences between RD and afterthoughts (Wei and Li 2017, 2018; Yip 2020). Crucially, only the afterthought chunk can be focused, as discussed in §2. Wei and Li (2017)'s and Yip (2020)'s acoustic experiments show that an afterthought chunk forms its own ι in Cantonese and Mandarin, as indicated by preceding pauses and initial pitch resetting—the chunk can even contain a boundary tone in Cantonese (see also fn.16). Hence, sluicing itself does not necessarily render prosodic rephrasing in both languages; rather, as we argue here, defocus does.

6.2 The degree of integration

The prosodic phrasing of RD has several possibilities even without mapping the RD chunk onto an ι . (94) represents different degrees of integration of the RD chunk into the main chunk. We argue that the first three possibilities (a–c), with a lower degree of integration, are illicit. Among them, the RD chunk is mapped onto a separate/maximal ϕ (abbreviated as **RD- ϕ** below), with differences in how RD- ϕ is parsed with the ι matching the main chunk CP1 (abbreviated as **CP1- ι** below). The remaining options (d–e) have a higher degree of integration, where the RD chunk does not form a separate ϕ , but part of some (maximal) ϕ . We argue that both are possible and the choice may be a matter of individual variations in grammar, as will be discussed below.

(94) Degree of integration (from low to high; RD chunk = italicized ZP)

- a. Unparsed: ((ZP) $_{\phi}$ (YP) $_{\phi}$) $_{\iota}$ (ZP) $_{\phi}$ *illicit*
 b. Recursive ι : (((ZP) $_{\phi}$ (YP) $_{\phi}$) $_{\iota}$ (ZP) $_{\phi}$) $_{\iota.max}$ *illicit*
 c. Separate ϕ : ((ZP) $_{\phi}$ (YP) $_{\phi}$ (ZP) $_{\phi}$) $_{\iota}$ *illicit*
 d. Recursive ϕ : ((ZP) $_{\phi}$ ((YP) $_{\phi}$ (ZP) $_{\phi}$) $_{\phi.max}$) $_{\iota}$ *the proposed one*
 e. Smaller than ϕ : ((ZP) $_{\phi}$ (YP ZP) $_{\phi}$) $_{\iota}$ *an alternative*

Possibilities (94a) with unparsed RD- ϕ and (94b) with recursive ι are empirically untenable: a right ι boundary between the two chunks is absent, as seen from §4–5. To rule out (94a), we add the PARSE constraint in (95), which requires all prosodic units in a sentence to be parsed into some ι (presumably the ι_{max}). We assume that this constraint outranks MATCH(CP, ι).

- (95) a. PARSE
 Assign a violation mark if Cⁱ is not dominated by ι , where Cⁱ $\neq \iota$.
 b. PARSE » MATCH(CP, ι)

In (96), the higher-ranked PARSE eliminates the unparsed candidate (b). Candidate (a), where the RD chunk is parsed with CP1 altogether as one ι , is the optimal candidate.

(96) Against unparsed RD chunks

[CP ₁ ZP YP] _i [CP ₂ ZP _{Df} YP] _j	PAR	ι :H	DEFOC	MATCH(CP, ι)
 a. ((ZP) $_{\phi}$ ((YP) $_{\phi}$ (ZP) $_{\phi}$) $_{\phi.max}$) $_{\iota}$				**
b. ((ZP) $_{\phi}$ (YP) $_{\phi}$) $_{\iota}$ (ZP) $_{\phi}$	*!			*

Another possibility (94b) integrates the unparsed RD- ϕ into CP1- ι , forming a recursive ι_{max} . To rule it out, We propose that Cantonese and Mandarin align their heads of ι with the right edges of ι , as in (97) (assuming categorical align, McCarthy 2003). Support from Cantonese comes from boundary tones, which are always at the right edges of ι , but not, say, the left edges (see Wong, Chan, and Beckman 2005's C-ToBI in Table 3). Support from Mandarin comes from the nuclear stress assignment rule proposed by Feng (2019, 65), which assigns nuclear stress to the rightmost elements in the VP in ι , explaining a number of word order alternation phenomena, also applicable to Cantonese.³⁴ Crucially, We suggest that ALIGN(ι ,RIGHT,HEAD(ι),RIGHT) (AL- ι -R) is ranked *higher* than MATCH(CP, ι), such that head directionality is more important than the mapping constraint on the ι -level.

- (97) a. ALIGN(ι ,RIGHT,HEAD(ι),RIGHT), abbreviated as AL- ι -R
Align the right edge of each intonational phrase with the right edge of its head Head(ι).
b. AL- ι -R » MATCH(CP, ι)

The tableau in (98) shows the attested candidate (a) without ι -recursion (=94d), and three non-optimal candidates (b-d) of (94b), where RD- ϕ and CP1- ι are grouped into a maximal ι_{max} . If RD- ϕ is not the head of ι_{max} as in (b), it yields a fatal ι :H violation. If ι_{max} is right-headed and RD- ϕ is the head as in (c), meaning that the DfP heads a higher prosodic category, a fatal DEFOC violation arises. If the ι_{max} is left-headed with CP1- ι as the head (which in turn has YP as the head) as in (d), AL- ι -R is violated. Candidate (d) ensures a match between CP1 and its corresponding ι and violates MATCH(CP, ι) only once, unlike (a) with two violation marks. Nevertheless, the ranking of AL- ι -R » MATCH(CP, ι) chooses (a) as the optimal candidate.

- (98) Against ι -recursion

[CP1 ZP YP] _i [CP2 ZP _{DfP} YP] _j	PAR	ι :H	DEFOC	AL- ι -R	MATCH(CP, ι)
 a. ((ZP) _{ϕ} ((YP) _{ϕ} (ZP) _{ϕk}) _{ϕ} . ι_{max}) _{ι}					**
b. (((ZP) _{ϕ} (YP) _{ϕ}) _{ι} (ZP) _{ϕk}) _{ι} . ι_{max}		*!			*
c. (((ZP) _{ϕ} (YP) _{ϕ}) _{ι} (ZP) _{ϕk}) _{ι} . ι_{max}			*!		*
d. (((ZP) _{ϕ} (YP) _{ϕ}) _{ι} (ZP) _{ϕk}) _{ι} . ι_{max}				*!	*

The remaining three possibilities in (94) parse the whole sentence into one ι , but differ in whether RD- ϕ forms a separate, maximal ϕ on its own like (94c), or it does not correspond to a maximal ϕ as in (94d–94e). Let us consider (94c), which does not involve ϕ -recursion, unlike the proposed (94d):

- (94c) Separated ϕ : ((ZP) _{ϕ} (YP) _{ϕ} (ZP) _{ϕ}) _{ι} *illicit*
(94d) Recursive ϕ : ((ZP) _{ϕ} ((YP) _{ϕ} (ZP) _{ϕ}) _{ϕ} . ι_{max}) _{ι} *the proposed one*

In order for (94c) to satisfy the higher-ranked ι :H and DEFOC, ι cannot be headed by the rightmost RD chunk, but needs to be either left- or middle-headed, violating right-headedness of ι . To maintain right-headedness, we suggest that head directionality (i.e., AL- ι -R) is weighed heavier than a constraint on recursion, such as NoRECUR in (99). This rules out left-/middle-headed ι in candidates (d) and (e)

34. Feng's Nuclear Stress Rule differs from English in including the notion of VP, such that SFPs, as prosodically weak elements (clitics), do not receive nuclear stress. Alternatively, one can take the Nuclear Stress Rule in Mandarin as the same as in English, with the specification that only categories larger or equal to ϕ may receive nuclear stress. Since SFPs do not form ϕ on their own, they also do not receive nuclear stress, and cannot be the head of ι .

in (100), and integrates RD- ϕ into a larger maximal ϕ as in candidate (a).^{35,36} Candidates (b) and (c) are ruled out by the now familiar ι :H and DEFoC violations.

- (99) a. NoRECUR (after Selkirk 1996)
Assign a violation mark if C^i dominates another C^i .

- b. AL- ι -R » NoRECUR

- (100) Against left-/middle-headed ι

$[CP_1 \text{ ZP YP}]_i [CP_2 \text{ ZP}_{DfK} \text{ YP}]_j$	ι :H	DEFoC	AL- ι -R	NoRECUR	MATCH(CP, ι)
☞ a. $((\text{ZP})_\phi ((\text{YP})_\phi (\text{ZP})_{\phi k})_{\phi \text{max}})_\iota$				*	**
b. $((\text{ZP})_\phi (\text{YP})_\phi (\text{ZP})_{\phi k})_\iota$	*!				**
c. $((\text{ZP})_\phi (\text{YP})_\phi (\text{ZP})_{\phi k})_\iota$		*!			**
d. $((\text{ZP})_\phi (\text{YP})_\phi (\text{ZP})_{\phi k})_\iota$			*!		**
e. $((\text{ZP})_\phi (\text{YP})_\phi (\text{ZP})_{\phi k})_\iota$			*!		**

The final possibility is (94e) (reproduced below), where the RD chunk forms some unit smaller than ϕ , presumably a prosodic word ω , violating a mapping constraint that a syntactic phrase (i.e., DeFoCp) should correspond to ϕ (i.e., MATCH(XP, ϕ) in Selkirk 2011).

- (94e) Smaller than ϕ : $((\text{ZP})_\phi (\text{YP ZP})_\phi)_\iota$ *an alternative*

While both Cantonese and Mandarin have no phonological process to differentiate between ω and ϕ boundaries, prosodic cues like pitch reset are helpful in this case. The degree of pitch reset at the left edges of ω (across words) is different from that of ϕ (across phrases) (see Yang and Wang 2002 for Mandarin; G. Li 2017 for Cantonese). Recall that from §5, we observed individual variations in the extent of pitch reset: some have a slight intermediate pitch reset, some have none. Hence, we suggest that both (94d) and (94e) are possible parses: some individuals may employ a “deeper integration” strategy and group the RD chunk as a ω instead of a non-maximal ϕ . To capture the difference formally, we propose that those speakers with some pitch reset, called Group I, rank the mapping constraint MATCH(XP, ϕ) in (101) higher than NoRECUR. Other speakers with no pitch reset, called Group II, penalize recursion by ranking NoRECUR higher than MATCH(XP, ϕ). The rankings are given in (101).

- (101) a. MATCH(XP, ϕ) (after Selkirk 2011)
Assign a violation mark if the left and right edges of a XP (non-clause and non-word) in the input syntactic representation do not correspond to the left and right edges of an phonological phrase in the output phonological representation.
- b. Group I speakers: MATCH(XP, ϕ) » NoRECUR
Group II speakers: NoRECUR » MATCH(XP, ϕ)

35. Empirically, it is difficult to differentiate a maximal ϕ boundary from a non-maximal ϕ boundary in Cantonese and Mandarin, which have no downstepping like the Japanese sort (Ito and Mester 2012, 2013). A potential way is to contrast the degree of pitch reset between maximal and non-maximal ϕ boundaries. Yet, it requires a very careful experimental design with ω boundaries also taken into considerations (i.e., possibility (94e), as discussed below). We leave this issue to future research.

36. For the headedness on the ϕ -level, we assume that ϕ can be left-headed (i.e., by YP in (100a)) in Cantonese and Mandarin, such that the RD- ϕ does not receive prominence on the ϕ -level as well (see, for example, Duanmu 2014 for left-headed phrasal stress in Mandarin).

The relevant results are shown in the tableaux (102) (assuming ϕ to be left-headed, see fn.36): Group I speakers parse the RD chunk as a non-maximal ϕ , whereas Group II speakers parse it as a unit small than ϕ , namely, a prosodic word ω .³⁷ In this way, the individual variations in pitch reset stem from the differences in individual phonological grammars.

(102) a. ϕ recursion (**Group I**)

$[_{CP1} ZP YP]_i$ $[_{CP2} ZP_{Df\phi} YP]_j$	$\iota:H$	DEFOC	M.(CP, ι)	M.(XP, ϕ)	NoRECUR
☞ a. $((ZP)_\phi ((YP)_\phi (ZP)_{\phi k})_{\phi.max})_\iota$			**		*
b. $((ZP)_\phi (YP_\omega ZP_\omega)_\phi)_\iota$			**	*!*	

b. Reduction to ω (**Group II**)

$[_{CP1} ZP YP]_i$ $[_{CP2} ZP_{Df\phi} YP]_j$	$\iota:H$	DEFOC	M.(CP, ι)	NoRECUR	M.(XP, ϕ)
a. $((ZP)_\phi ((YP)_\phi (ZP)_{\phi k})_{\phi.max})_\iota$			**	*!	
☞ b. $((ZP)_\phi (YP_\omega ZP_\omega)_\phi)_\iota$			**		**

Taking stock, we have proposed an Optimal-Theoretic analysis that derives the syntax-prosody mismatches in RD by stressing the role of defocus. Combining with headedness on the ι -level, defocus disallows the RD chunk, being a CP, to be mapped onto an ι . We have also discussed different extents of integration of the RD chunk into the main chunk and argued in favor of a “deep integration” strategy, where the RD chunk does not constitute an independent maximal ϕ and is parsed into some ϕ in the main chunk. In the next section, we zoom out to the broader picture of focus in the syntax-prosody mapping and discuss the role of defocus as compared to focus.

7 Focus and defocus in syntax-prosody mapping

As briefly mentioned in the introduction, the approaches on focus in syntax-prosody mapping may be characterized into two groups: the REPHRASING view and the NO-REPHRASING view. Below, we first review the two approaches, and then discuss the implications of the current DEFOCUS REPHRASING view on the syntax-prosody mapping.

In the REPHRASING view (Pierrehumbert and Beckman 1988; Truckenbrodt 1995; Selkirk 2008, *i.a.*), focus may indirectly trigger rephrasing via edge alignment, as schematized in (103). We will illustrate the details of the REPHRASING view using Truckenbrodt (1995)’s system, and discuss which component is subject to debate in the NO-REPHRASING view.

(103) Focus in the REPHRASING view

Focused constituent
 $\downarrow$ FOCUS
 Metrical prominence
 $\downarrow$ *mapping axiom*
 Prosodic head
 $\downarrow$ ALIGN
 Edge of a prosodic constituent

37. An alternative parse $((YP)_\phi ZP_\omega)_{\phi.max}$, where YP forms a ϕ on its own and combines with ZP to form a recursive ϕ , violates both MATCH(XP, ϕ) (once) and NoRECUR (once) and is not possible.

First, focused elements (focus) are assigned metrical prominence, via a constraint Focus in (104). Focused elements (F) contain the highest metrical prominence in the domain of focus (DoF), namely, the (semantic) scope of the focus where the background is contained (Rooth 1992). DoF is often the whole sentence in single-focus cases.³⁸

- (104) Focus (adapted from Truckenbrodt 1995:11; cf. Jackendoff 1972:237)
If F is a focus and DoF is its domain, then the highest prominence in DoF will be within F.

Second, bearing metrical prominence, focus is the head of the prosodic domain it occurs in. This is achieved by an axiom in (105) that maps elements with metrical prominence in a domain to the prosodic head of that domain. This axiom is inviolable in Truckenbrodt (1995)'s framework (though challenged by Ishihara 2011). For expository purposes, we refer to this mapping axiom as MAP(P,H) (i.e., mapping prominence to head).

- (105) MAP(P,H): One-to-one mapping between metrical prominence and prosodic head
(as an inviolable axiom, Truckenbrodt 1995:12)
Each grid-mark in the metrical representation of prominence is the unique head of a constituent in the prosodic representation.

For example, when a word ω bears focus, it becomes the head of the phonological phrase ϕ (notated by underline), as in (106). In other words, *focus* prominence is mapped onto *head* prominence. This captures the associated phonetic effects on the focal field such as a rise in f0 peak.

- (106) a. $[[_{XP} \omega_1 \omega_{2F}] [_{YP} \omega_3]]_{DoF}$
b. $(\omega_1 \underline{\omega_{2F}})_{\phi_1} (\underline{\omega_3})_{\phi_2}$

Third, as a head, focus is aligned to the edges of the focus domain, such as the sort of ALIGN(ι ,R,HEAD(ι),R) in (97). A generalized ALIGN constraint is given in (107). If focus originates in the middle of a prosodic domain, rephrasing such as boundary insertion or deletion may occur to align the focus to the edges, constituting a way where prosodic rephrasing may mismatch with syntactic structure, as schematized in (108) (assuming right-headed ι).

- (107) ALIGN (Truckenbrodt 1995:12)
If Pros is a prosodic constituent, let Head(Pros) be the grid mark that heads Pros in accord with (105). Then: Align(Pros, L, Head(Pros), L) or Align(Pros, R, Head(Pros), R)
i.e., Align the left/right edge of each prosodic constituent Pros with the left/right edge of its head Head(Pros).

- (108) Boundary insertion (boldface) and deletion (strikethrough)
a. $[[_{XP} \omega_1 \omega_{2F}] [_{YP} \omega_3]]_{DoF}$ b. $((\omega_1) (\underline{\omega_{2F}})_{\phi_1} (\underline{\omega_3})_{\phi_2})_{\iota}$
* $((\omega_1 \underline{\omega_{2F}})_{\phi_1} (\omega_3)_{\phi_2})_{\iota}$ $\rightarrow ((\omega_1)_{\phi_1} (\underline{\omega_{2F} \omega_3})_{\phi_2})_{\iota}$

38. The domain of focus is calculated according to Rooth (1992) alternative focus system. In other words, DoF is based on the semantic scope of focus, which yields the alternative set that contains the focus and the background. For example, in *Only John eats lamb*, the alternative set is {John eats lamb, Mary eats lamb, Bill eats lamb}. The scope of *only* here is thus the whole preajacent clause, *John eats lamb*, which is the domain of focus. The focus domain DoF is by default the clause where the focus occurs in, but it varies with the scope of the focus. In (i), for instance, when two contrastive foci are coordinated, the DoF of each focus is each conjunct, instead of the whole sentence.

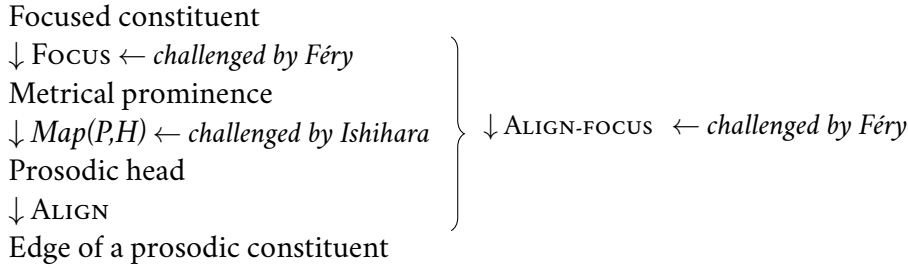
(i) [An [American]_F farmer]_{DoF} and [a [Canadian]_F farmer]_{DoF} went to a bar. (Truckenbrodt 1995:161)

- c. [VP [PP Mari **ni**_F] [V atta]]
 (Mari) (**ni**_F atta)
 Mari with met

(Japanese, Truckenbrodt 1995:212)

However, the REPHRASING view has been recently argued to be too strong, and an alternative NO-REPHRASING view is proposed (Féry and Ishihara 2010; Ishihara 2011, 2016; Féry 2013; see also Wagner and McAuliffe 2019; D. Wu 2021), as schematized in (109). Center to their point is that focus prominence does not *have* to be aligned with prosodic edges.

(109) Focus in the NO-REPHRASING view



Ishihara (2011, 2016), based on Japanese, a major language that motivates Truckenbrodt's system, argues that MAP(P,H) is a violable constraint, reformulated as CULMINATIVITY in (110). He shows that Japanese focus does not block downstepping, meaning that focus does not trigger rephrasing and create a new prosodic domain for downstepping. For him, focus may directly manipulate the pitch register (see Wagner and McAuliffe 2019 for possible models), and there is no need to attribute focus prominence to head prominence. A prosodic unit (ϕ and ι) can well contain two prominent members, the focus and the head. In other words, there can be a dissociation of (focal) prominence and head, resulting in a dissociation of (focal) prominence and edge.

- (110) CULMINATIVITY (as a violable constraint) (Ishihara 2011:1881)
 Each prosodic phrase (ϕ , ι) contains one and only one grid-mark (i.e., prominence).

Féry (2013) also objects to the idea that focus must be aligned with edges, by replacing MAP(P,H) and ALIGN with a set of *violable* ALIGN-FOCUS constraints in (111).³⁹ She shows that alignment is a preference and is not observed in a wide range of languages like French, Northern Bizkaian Basque, Konkani (Indo-Aryan), German and Fon (Kwa, Gbe). Moreover, she argues against Focus by considering languages where focus is not marked by metrical prominence like French. In other words, both focal prominence and edge, and focus and metrical prominence, are dissociated.

- (111) ALIGN-FOCUS {L/R}, { ϕ / ι }-PHRASE {L/R} (as a violable constraint) (Féry 2013:691)
 Align a focus with the {left/right} boundary of an {prosodic/intonational} phrase.

With the theoretical background in place, we now turn to the current proposal regarding defocus. We suggest that defocus should also be dissociated from focus in the syntax-prosody mapping. We first discuss the roles of defocus and focus, and then the realization of prosodic prominence.

First, recall that focus does not trigger rephrasing in both Cantonese and Mandarin. Cantonese does not mark focus prominence by f0 cues, but usually by durational cues (Wu and Xu 2010; Fung and Mok 2014; Hsu, Xu, and Ngai 2018; Yang 2022, *pace* Man 1999; Gu and Lee 2007), which are even

39. Four constraints are represented by (111): (i) ALIGN-FOCUS R, ι -PHRASE R, (ii) ALIGN-FOCUS L, ι -PHRASE L, (iii) ALIGN-FOCUS R, ϕ -PHRASE R, and (iv) ALIGN-FOCUS L, ϕ -PHRASE L.

not obligatory (see C. Lee 2024 for group variations). Importantly, Cantonese does not have post-focal reduction, regardless of the types of phonetic cues (Gu and Lee 2007; Wu and Xu 2010; W. L. Wu 2013; B. Li 2024). To the extent that post-focal reduction is absent, we can safely conclude that Cantonese focus does not affect the phrasing of surrounding environments; in other words, there are no rephrasing effects regardless of the position of the focus.

Mandarin, on the other hand, has both focal prominence (e.g., boosted f0 peak and enlarged f0 range) and post-focal reduction (e.g., lowered f0 and reduced f0 range) (Chen, Wang, and Xu 2009; Xu, Chen, and Wang 2012; Xu, Xu, and Sun 2004), which at first glance seems like a language that has focus-rephrasing effects. It has, however, recently been uncovered that phrasing effects are retained in post-focal fields in Mandarin, suggesting that the post-focal reduction should not be attributed to boundary deletion along Truckenbrodt's lines (Zhang, Clayards, and Wagner 2024). Note that this is a case similar to English (Wagner and McAuliffe 2019; D. Wu 2021) and Japanese (Ishihara 2011, 2016), where focus receives prominence but does not trigger rephrasing. Hence, the core idea of the NO PHRASING view should apply to Mandarin, as well as Cantonese: focus does not necessarily align with prosodic edges. That is, MAP(P,H)/CULMINATIVITY (or ALIGN-FOCUS) should be a violable constraint and ranked lower than the syntax-prosody mapping constraints in both languages:

- (112) Focus-head/focus-edge alignment constraints are ranked low in Cantonese and Mandarin
 $\{\text{MATCH}(\text{CP}, \iota), \text{MATCH}(\text{XP}, \phi)\} \gg \{\text{MAP}(\text{P}, \text{H})/\text{CULMINATIVITY}, \text{ALIGN-FOCUS}\}$

Since defocus triggers rephrasing but focus does not, a dissociation between the two is necessary. In OT terms, defocus-related constraints in both languages are ranked higher than focus-related constraints, such that it is more costly to violate the defocus ones than the mapping constraints, but less so to violate the focus ones, as represented by (113). Put differently, defocus is *not* merely the complement of focus and its effects on mapping cannot be derived by focus. It should be regarded as its own category. This is the core idea underpinning the current DEFOCUS REPHRASING view.

- (113) Defocus vs. focus in the syntax-prosody mapping in Cantonese and Mandarin
 $\overbrace{\{\iota: \text{H}, \text{DEFOC}\} \gg \{\text{MATCH}(\text{CP}, \iota), \text{MATCH}(\text{XP}, \phi)\}}^{\text{defocus triggers rephrasing}} \gg \underbrace{\{\text{MAP}(\text{P}, \text{H})/\text{CULMINATIVITY}, \text{ALIGN-FOCUS}\}}_{\text{focus does not trigger rephrasing}}$

The dissociation of defocus from focus predicts that non-focused elements, even occurring post-focally, are distinct from defocus. Since focus does not trigger rephrasing, non-focus may still be a prosodic head with some prominence, as compared to defocus which loses prominence altogether. This possibility is schematically represented in (114), where a focus (XP) in the middle of CP1 is followed by non-focused elements (YP) in CP1 and defocus (ZP) in CP2. While defocus triggers rephrasing and forms a unit with the preceding YP, the focus forms its own ϕ and does not, say, erase the post-focal ϕ boundaries.

- (114) a. $[\text{CP1}_{\text{main}} \text{ZP}_{\text{Non-focus}} \text{XP}_{\text{Focus}} \text{YP}_{\text{Non-focus}}] [\text{CP2}_{\text{RD}} \text{ZP}_{\text{Defocus}}]$
 b. $((\text{ZP}_{\text{Non-focus}})_{\phi} (\text{XP}_{\text{Focus}})_{\phi} ((\text{YP}_{\text{Non-focus}})_{\phi} (\text{ZP}_{\text{Defocus}})_{\phi})_{\phi_{\text{max}}})_{\iota}$

While our experiments were not designed to distinguish nonfocus from defocus (e.g., the contexts were intended for broad focus in main chunks), several Mandarin participants produced a narrow focus in the main chunk in some experimental items of RD, such as the gapless RD in (115). The modal *gai1* 'should' is focused, as can be seen from its f0 boost and the post-focal reduction on the

following elements *kai1 can1ting1* ‘open a restaurant’ in the main chunk in Figure 8.

(115) Narrow focus placement in dislocation copying in Mandarin

Xiao1bing1 dong1tian1 **GAI1** kai1 can1ting1 ma0 Xiao1bing1?

Xiaobing winter should open restaurant SFP Xiaobing

‘SHOULD Xiaobing open a restaurant in the winter, Xiaobing?’

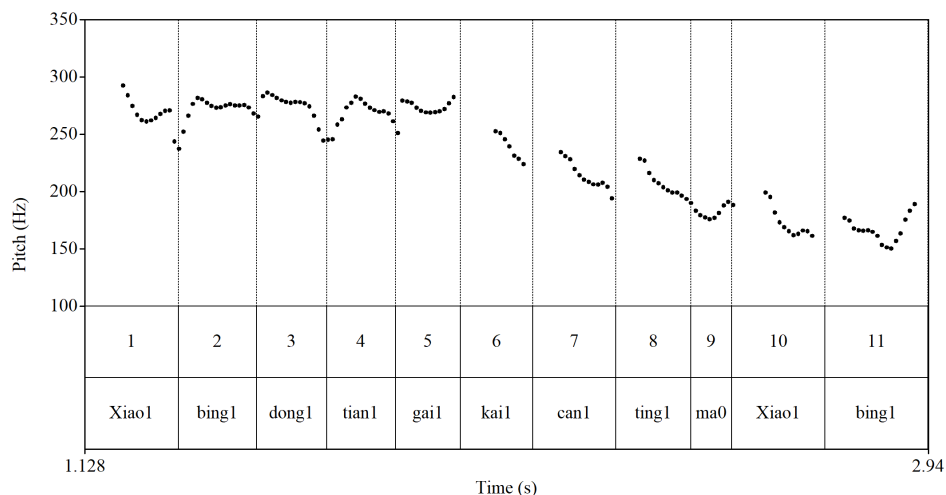


Figure 8: f0 tracks of the gapless RD sentence (115) in Mandarin (M01)

While the post-focal elements are “compressed” with lower f0 (σ_6 - σ_8), they are still more prominent than the defocus elements, which have a much lower f0 (σ_{10} - σ_{11}). In other words, the post-focal reduction effects are different from and independent of defocus rephrasing.⁴⁰ If both of them were achieved by the same mechanism such as boundary deletion all the way to the end of the sentence, one should not see a difference in prominence between post-focal elements and defocus. On the other hand, if one adopts the NO-REPHRASING view where focus may alter the pitch register directly, the difference between post-focal elements and defocus can be explained: the former results from direct manipulation by focus, and the latter results from prosodic phrasing (non-head). This offers strong support for the dissociation between defocus and focus.

To summarize this section, we have argued for a DEFOCUS REPHRASING view, where defocus plays a distinct role from focus in the syntax-prosody mapping, and their effects on prosody should be distinguished, and indeed can be distinguished empirically. The implications from the current proposal is that we should weaken the current REPHRASING view by dissociating prominence with various notions as suggested by the NO REPHRASING view, including (i) focus, (ii) headedness, and most importantly, (iii) non-focus and defocus. The novelty of the current view lies on the third one, where non-prominence resulting from *defocus* may trigger rephrasing, but non-prominence resulting from post-focal reduction does not, hence representing a weaker REPHRASING view.

40. One may object to this conclusion and treat the difference in f0 as natural declination, found in Mandarin declaratives (e.g., Xu 1999). Yet, the sentence (115) in Figure 8 is interrogative, and interrogative sentences (with canonical word order) in Mandarin have exponential global rising towards the end (Liu and Xu 2005), instead of declination. Of course, a follow-up experiment should be conducted in the future to fully examine the phonetic difference between non-focus and defocus.

- (116) A **DEFOCUS REPHRASING** view that distinguishes between:
- a. **Focus** vs. **prominence** (such that focus does not always have prosodic reflex);
 - b. **Focus** prominence vs. **Head** prominence (such that focus does not always trigger rephrasing);
 - c. Non-prominence resulting from **non-focus** (i.e., post-focal reduction) vs. Non-prominence resulting from **defocus** (i.e., non-head).

8 Concluding remarks

To conclude this paper, we have investigated the understudied effects of defocus on syntax-prosody mapping through the lens of right dislocation in Cantonese and Mandarin with independent evidence from each module of grammar. Recall the three research questions:

- (117) a. Is there a syntax-prosody mismatch in RD in Cantonese and Mandarin?
 b. If yes, does defocus play a role in triggering the mismatch?
 c. Do the findings support a dissociation of the roles of defocus and focus in prosody?

To answer the first question, we demonstrated that syntactically, RD has two clauses, based on a novel typological argument of the correlation between empty category inventories and availability of gapped RD, as well as a series of language-internal arguments in the literature. Drawing evidence from phonological processes like boundary tone placement in Cantonese and third tone sandhi in Mandarin, we showed that RD nevertheless has only one intonational phrase. The prosodic phrasing is further supported by phonetic cues examined by acoustic experiments, such as the lack of final lengthening, full pitch reset, and pauses within RD. There is hence a syntax-prosody mismatch.

As for the second question, semantically, RD chunks systemically resist alternative-based focus (which is different from discourse newness/givenness), and hence are always defocused. We argued that defocus, being the mirror image of focus, cannot receive prosodic head prominence. Therefore, it cannot be a prosodic head, which, in the case of RD, leads to an illegitimate headless ι . To avoid headless prosodic constituents, the syntax-prosody mapping “compromises”, modeled Optimality-Theoretically as a lower ranking of $\text{MATCH}(\text{CP}, \iota)$ than $\iota:\text{HEAD}$ and **DEFOCUS**. The defocused RD chunk is thus parsed with the preceding main chunks as one ι , leading to the syntax-prosody mismatch.

In relation to the third question, focus in both languages importantly does not trigger rephrasing (e.g., Wu and Xu 2010; Zhang, Clayards, and Wagner 2024), suggesting an asymmetry between focus and defocus. The rephrasing effects in RD cannot be attributed to non-prominence due to post-focal reduction (non-focus), but rather a genuine result of the lack of head prominence (defocus). We thus proposed a **DEFOCUS REPHRASING** view, where defocus is dissociated from focus in syntax-prosody mapping, as represented as a lower ranking of focus alignment/phrasing constraints (e.g., **ALIGN-FOCUS**) than $\text{MATCH}(\text{CP}, \iota)$ or $\text{MATCH}(\text{XP}, \phi)$. Effectively, this allows a language to have defocus-rephrasing effects without having focus-rephrasing effects, which is precisely the case in Cantonese and Mandarin. It implicates that focus vs. prosodic head should be further differentiated in prosodic prominence, and post-focal fields vs. defocus/non-heads should also be differentiated in non-prominence. This implication resonates with the emerging idea that different prosodic representations should be kept separate (Ishihara 2011, 2016; Wagner and McAuliffe 2019).

As a final remark, RD is empirically diverse cross-linguistically, in terms of both focus interpretation and prosodic phrasing. On one hand, RD in several languages is argued to allow a focus element, such as Japanese (Yamashita 2011; Takita 2011; Abe 2019), Korean (Ko 2015), and Mongolian (T. T.-M. Lee 2023). Whether their prosodic organization is correlated with the focus occurrence remains to be explored, but some preliminary results from Nakagawa, Asao, and Nagaya (2008) suggests the degree of integration of RD chunks to main chunks in Japanese varies with the presence/absence of focus. On the other hand, it has been reported that RD in some languages like French (Delais-Roussarie, Doetjes, and Sleeman 2004; Ladd 2008) and Catalan (Feldhausen 2010) involves a bi-intonational phrase structure, but their RD also resists focus (Lambrecht 1981; Vallduví 1994), like Cantonese and Mandarin. This suggests that the syntax-prosody mapping in French and Catalan RD is isomorphic, a variation that is expected under the OT framework where constraint ranking is specific to individual languages. While there remain a number of interesting questions to be answered, what is important about the current DEFOCUS REPHRASING approach is that it predicts a potential factorial typology that can be examined by these cross-linguistic variations, and hence allows us to study the topic in a theoretically-informed manner.

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